SAINT MICHAEL COLLEGE OF CARAGA

SMCC COMMUNITY EXTENSION SERVICE (CES) WORKFLOW SYSTEM WITH AI ANALYSIS AND RECOMMENDATION

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CHAPTER 1

INTRODUCTION

1.1 Project Context

The Community Extension Services (CES) of Saint Michael College of Caraga has been instrumental in connecting the school to the community. Through numerous programs and activities, CES has made a significant impact on people's lives. Its primary goals are to help people become stronger, improve their quality of life, teach them to care for the environment, and protect their culture. These activities not only benefit the community but also provide students with the opportunity to apply what they learn in school in real-life situations. Some of the programs done by CES are Computer Literacy Training for barangay officials in Barangay Camagong, Nasipit; Paskong Michaelinians for the elderly at Por Cristo in Buenavista; medical check-ups and blood donation drives; tree planting in Mount Kasunugan and other areas; and skills training for mothers so they can earn extra income for their families. These are just some of the many projects that continue to help and support the community.

By 2025, however, the many CES activities also caused problems. Although modern technology can make tasks easier, reports are still mostly completed on paper. Reports still need to be done in person and organized manually. This often leads to late submissions, difficulty finding old reports, and the loss of important documents. As a result, staff spend more time on paperwork than on improving community programs.

Technology refers to the application of knowledge and skills to design, develop, and implement tools, machines, or systems that facilitate problem-solving and streamline work processes [1]. For CES, this means using digital tools can make reporting faster, reduce mistakes, and better manage reports

and activities. Research also shows that paper-based offices require substantial storage space and ongoing organization to maintain efficiency.

The paper industry also adds to pollution, deforestation, and waste [2] . This contradicts CES's values, which emphasize the importance of caring for people and the environment. That is why moving to digital systems is important.

Studies also show that Artificial Intelligence (AI) is now utilized in many organizations to aid in prediction, support decision-making, and provide valuable insights [3]. For CES, this means AI can be utilized to analyze reports, offer recommendations, and enhance the effectiveness and efficiency of services.

Without a proper system, it is challenging to monitor activities, identify the most successful programs, and pinpoint areas for improvement. Coordinators and school leaders also find it challenging to work together, as reports are often submitted manually or through basic online platforms without a clear structure. This results in delays, lost information, and missed opportunities to utilize data for improved planning. Without AI recommendations based on past reports, CES also loses opportunities to improve future projects.

Due to these challenges, the researchers proposed the SMCC Community Extension Service (CES) Workflow System, incorporating AI Analysis and Recommendation. This System provides a simple online platform where coordinators can create and submit reports, while administrator can review, approve, or reject them. It also features AI capabilities to verify data accuracy and offer recommendations for future projects. By reducing paper use, keeping reports safe and organized, and enabling leaders to make smarter decisions, the System makes CES more efficient, environmentally friendly, and more beneficial to the community.

1. 2 Literature Review

This displays the relevant literature from the researchers' investigation, aligned with the issues currently being faced. It emphasizes the need to present the researchers' suggested software program, its advantages, and the problems it resolves. Literature originates from various sources, including websites, journals, papers, e-books, and electronic materials.

Community Extension and Local Development

Since beneficiaries and adopters participated in communal efforts such as HELP and KKK, the Nueva Vizcaya State University College of Teacher Education's extension services were also beneficial. Through such extension services, beneficiaries achieved significant improvements in their quality of life through livelihood, health, and literacy interventions. Since such services seemed inspiring and relevant, they also inspired communal change. The total value achieved by the extension services was an indicator of its success and relevance to the institution and community, despite a few minor complications, such as erratic consultations and monitoring issues [4].

Through community extension projects, the research highlights how institutions of higher learning can combat poverty by fostering capacity development and entrepreneurship training. It shows that access to credit centers, budgeting, financial literacy, and entrepreneurship training significantly reduce poverty and empower communities to improve their socioeconomic futures [5].

According to studies, community extension services provided by tertiary institutions play a significant role in enhancing people's social well-being and alleviating poverty. Inclusive literacy, health, and livelihood interventions, as well as group programs like KKK and HELP, have significantly improved the lives of beneficiaries and instilled social change. Meanwhile, programs on financial literacy,

budgeting, and entrepreneurship education, as well as access to credit facilities, enable people to achieve sustainable livelihoods and a more stable socioeconomic future. These extension efforts demonstrate institutional relevance and transformative effects on societies, despite minor issues with tracking and consultation. The small sample size in the study—the combined total of 90 respondents across the three villages—will limit the extent to which the findings can be generalized.

This study serves as a disadvantage for the Community Extension. Furthermore, since it focuses almost exclusively on Mount Merbabu National Park, the findings will be place-specific rather than being universally applicable. In addition, if social capital and human capital are the focus of the study, other potential factors that can determine people's community involvement in conservation efforts are left unexplored [6].

The research is relatively small, with only 26 family beneficiaries, which significantly limits the scope of the conclusions and their broader applicability to a larger population. Additionally, as the research is based on a single community, the findings, being context-specific, exhibit a significant lack of generality across the entire scope [7].

The evidence clearly shows that the primary disadvantage of community extension research is its narrow scope and emphasis on context-specific factors. The outcome derived from Mount Merbabu National Park remains inherently place-specific, as it focuses solely on the facets of human and social capital without considering other important factors that are truly efficacious for community engagement. Similarly, research that relies on only 26 family beneficiaries in a specific town undermines the generalizability of its findings. These constraints complicate the extrapolation of evidence to larger and more diverse populations, thereby highlighting the substantial lack of broad applicability in this field of research.

Digital Workflows and Administrative Efficiency

Community extension programs can benefit from the systematic Management Information System (MIS) in a variety of ways. With the automation of recruitment, monitoring, and data reporting, manual work decreases and processes accelerate, boosting efficiency. An improved monitoring system and clear digital records boost liability and transparency. Decision-making is also easier when based on actual data, allowing managers to plan based on real results. In addition, collaboration among stakeholders improves, and resources such as time, funds, and workforce are more effective. Ultimately, stakeholders show their readiness for digital transformation, and MIS is designed for long-term, sustained operation [8].

According to [9], the design and installation of the University Management Information System have different benefits. This increases productivity by automating the institution's document processing and activities. It also ensures epic management by properly storing data, limiting access, and strengthening transparency. The System can also be adapted to the administration's needs and teaching practices due to its multi-module structure, which covers a variety of functions. Ultimately, it enhances the management of the learning process through automation and strengthens the university's decision-making capabilities. Stakeholder collaboration improves, and resources—time, funds, and workforce — are used more effectively. Ultimately, stakeholders show their readiness for digital transformation, and MIS is designed for long-term, sustained operation.

According to [10], the use of automated guided vehicles (AGVs) in public hospitals offers several benefits. Due to strong union organizations and opportunities for reassignment, there is a possibility of reduced unemployment, improved health and safety, and reduced heavy manual work. Technology also provides limited opportunities for skill enhancement, which still contributes to better working

conditions. In total, these examples show how digital technology and automation can change employment conditions, transparency, and operational efficiency. Whether used in hospitals, universities, or community programs, the integration of automation technology and management information systems enhances decision-making, strengthens accountability, and streamlines processes.

Additional benefits include better resource management, improved health and safety, and readiness for digital transformation. Although some issues, such as limited work changes and skill upgrading, are maintained, the findings show that the digital solution effectively improves institutions and supports long-term maintenance.

The study taught some weaknesses of digitalization. Due to the lack of extensive internet access and digital skills, poor groups such as adults, low-educated, workers, unemployed, and low-income remain at risk. Additionally, the fear of unemployment persists due to the use of machines, which creates inequality and fear. The findings show that, despite the majority's positive outlook on digitalization, its benefits to all citizens remain limited due to skill gaps and limited access to spaces [11].

According to [12] , digital transformation in rural areas has many limitations. It blocks socio-cultural and technological problems, even though it has the potential to enhance sustainability. It also faces economic and regulatory problems, and some communities remain at risk of being left behind. In addition, change has different effects; some are positive, but it also has socioeconomic, environmental, and institutional downsides that can reduce its benefits if not properly managed. Although technology simplifies life and makes information more affordable, it can also have adverse effects when overstated, such as the danger to private life, cyberbullying, and physical and mental health problems [13].

In summary, the findings demonstrate that although digitalization has numerous benefits, it also has significant weaknesses. Little-privileged groups are particularly affected by uneven access and skills, while rural communities face technological, cultural, and economic challenges that impede their transformation. Additionally, the excessive use and uncontrolled exploitation of digital devices can harm privacy, social relationships, and overall health. Ultimately, the findings show that the benefits of digitalization are not evenly distributed and may be affected by misuse and institutional barriers.

Document Automation and Management

The use of comprehensive architecture for multi-team process automation has excellent benefits. By applying machine learning, real-time analytics, and job automation, it significantly enhances operations and reduces redundant data processing. With automated job scheduling, integrated data streams, and adaptive workflows, efficiency and user experience are improved, while security is ensured through robust cybersecurity features. In enterprise-wide verification, the data has proven to be more accurate, tasks are faster, and team coordination is better. This System also facilitates alignment among stakeholders, scalability, and flexibility, thereby creating a broader, more effective automation process [14].

Meanwhile, the use of the Document Management System (DMS) is causing a paperless office. It increases productivity within the organization, reduces paper use, simplifies repetitive processes, and facilitates document movement across departments. Furthermore, by supporting the processes that can be exacerbated and reducing excessive use of paper, it also strengthens sustainability [2]

According to one study, the benefits of developing and implementing the web-based electronic document management system (WBEDMS) are clear. It increases user productivity and document-handling efficiency, and solves the problems with manual filing. It gained high evaluation ratings of 95% accuracy, 99.20% usability, and 96.60% user satisfaction, showing its efficiency and fitness for application to organizations [15]

In addition, in evaluating the use of Electronic Document Management Systems (EDMS), it has been found that users' intention to use them is strongly influenced by performance expectancy and social influence. It demonstrates the importance of perceived usefulness and organizational support in promoting technology adoption. The findings demonstrate the significant potential of EDMS adoption in schools, as they account for 61% of users' adoption intentions [16].

In total, studies have shown that the use of electronic document management systems (EDMS) and automation frameworks brings significant benefits to organizations. Automation prevents unnecessary actions, enhances accuracy, and strengthens collaboration through machine learning, real-time analytics, and adaptive workflows. By simplifying document handling, reducing costs, and supporting green practices through paperless processes, EDMS and DMS technology are further strengthening productivity. The findings also emphasize that perceived utility and organizational support encourage the use and ensure the successful implementation of these systems in academic and professional environments.

The EDMS application at Bartın University also showed some limitations. Although Social Influence and Performance Expectancy helped us, effort expectancy has no apparent effect, indicating that usability is not a strong motivation. This indicates that, despite the organization’s support and

perceived usefulness, the user's acceptance may remain limited if the System's processes are not simplified [17].

According to Laguna State Polytechnic University, weaknesses in the use of EDMs were identified. While it has a positive effect on output, effectiveness, and collaboration, there are disadvantages, such as the need to adapt the System according to institutional needs and problems caused by the complex user interface. Without proper repairs and support, these issues limit usability and may prevent greater implementation [18].

Some limitations of EDMS development for businesses on a local area network are also discussed. The System requires extensive configuration, ongoing compliance maintenance, and adaptation to users, despite its efficiency and cost-reduction benefits. These factors can be challenging to implement, especially for organizations with limited technical capabilities.

Technology Integration in Education

According to [1], the study's findings demonstrate that technology integration in school management has significantly contributed to various aspects, including digital literacy, equal access for all students, and the effective use of technology by students and their parents. Furthermore, it strengthens community participation, contributing to a more responsive, inclusive teaching environment that benefits everyone involved. The results show that integrating technology into school management clearly enhances digital literacy, promotes equal opportunities, and fosters active participation by both students and parents. Such integration deepens community participation, creating a more inclusive and cohesive educational environment that benefits all stakeholders.

However, studies also have weaknesses, such as relying solely on controlled conditions in the review of missing-data methods, which may not account for real-world complexities, especially in correctional departments. Moreover, only two risk-assessment tools are included in the review, which may limit the generalizability of the results compared with other tools [19].

In general, the research's specific focus and the use of controlled environments limit the scope of the findings. However, they provide important insights into managing missing data in correctional risk instruments. This means that, to expand the applicability of the findings, more studies need to be conducted using a broader range of instruments and in real-world settings.

Artificial Intelligence in Knowledge and Learning Systems

The report provides perceptive commentary on why the automation of work and the alleviation of workload by artificial intelligence increase productivity. It highlights why upskilling and reskilling are effective solutions on the ground for tackling the problem being created by the incorporation of AI. A valuable resource for organizations in the throes of a technological shift, the report also identifies the benefits of transversal competencies as much as creative training approaches [20]. The paper provides a systematic framework for knowledge sharing in non-profit healthcare organizations with limited resources, highlighting how AI can assist. It uses the Resource-Based View, the Dynamic Capabilities Theory, and the Absorptive Capacity Theory to provide it with a theoretical foundation. It ensures a systematic mapping of evidence and identification of key gaps to develop AI-based solutions for non-profit healthcare organizations, using a PRISMA-ScR-anchored approach [2].

The research outlines the use of recommender systems that integrate artificial intelligence across numerous industries, including business, health, finance, and academia. It introduces the newest state-of-the-art methods that improve recommendation efficiency, interpretability, and customization. The research offers insightful explanations for improving intelligent recommendation systems by proposing several innovative strategies [21].

By focusing on data properties rather than recommender system methods, the study introduces a new paradigm. It highlights the direct impacts on fairness and accuracy caused by sparsity, skewness, popularity bias, and noise. By demonstrating how data constraints can significantly reduce the algorithm's effectiveness, the study generates insights that will help inform system architecture and data management research in the coming years [22]. The paper provides experimental confirmation of the reliability of Cooperative Awareness Messages in realistic V2X communications. It demonstrates its reliability for safety use cases and accident investigation, with high accuracy across key parameters such as speed, acceleration, orientation, and yaw rate. The study also shows that new car models can further enhance data accuracy, facilitating the development of intelligent transportation systems [23].

The paper conducts an in-depth analysis of 16 national AI strategies and offers an insightful discussion of how public participation is handled in governance. The paper finds a predominance of professionalism, efficiency, and competition, along with an underemphasis on participation mechanisms. The research contributes to a better understanding of the pitfalls of neoliberal governance of AI and identifies areas where policy development becomes complex or incomplete [24].

Consolidated research recognizes the capacity of artificial intelligence to transform across domains such as management, healthcare, recommendations, transportation, and organizational

efficiency. These include the following: speed, accuracy, customization, and knowledge sharing, along with their limitations, such as a lack of control, data constraints, and the need for workers to adapt. In summary, the ability of AI to analyze data is demonstrated by the use of various tools for analyzing and interpreting data. Its potential is significant because data is presented to the public in many ways, such as through engagement, training, policy-making, and data analysis.

Developing recommendations using ChatGPT shows that consumer responses depend significantly on the type of recommendation agent and the apparent accuracy of the information. Although consumers have been positively responding to a variety of options, acceptance decreases when there is doubt about the data's accuracy or when the options are compared to the person, which limits ChatGPT's effectiveness as a recommender [25].

The limits of artificial data generation are also evident in manufacturing. While it helps address privacy issues and data scarcity, it still depends on proper data validation and preprocessing, as false datasets can skew accuracy. It can also be impractical in some industries as it requires sufficient technical knowledge and careful use [26].

The weaknesses of machine learning (ML) are evident in the health sector as well. High-quality, large datasets are required to be effective, but doctors can be reluctant to trust automated results due to the "black-box" nature of complex algorithms. In addition, skilled personnel and durable infrastructure are needed for practical application, but they may not always be available in inferior health systems [27].

Meanwhile, Big Data is rapidly changing the way we think and make human decisions, thanks to its ability to enhance discovery and decision-making. However, continuous data growth is no longer adequately addressed by traditional technology. Although the use of ML has expanded for specific applications, problems such as computational complexity, classification errors, and data heterogeneity remain. The study highlights the importance of ML in Big Data Analysis, its challenges, and its capabilities to provide valuable insights for predictive analysis and real-time decision-making [28].

Another study also highlighted the importance of machine learning in addressing cybersecurity problems arising from digitalization and the Internet of Things. Cyberattacks continue to rise, and traditional security measures are unable to keep pace. By leveraging cyber insights, cybersecurity becomes more dynamic, automated, and proactive through machine learning. The paper provides a deeper understanding of ML algorithms for intelligent data automation and analysis, discusses real-world applications, and suggests potential areas for future research [29].

Also, the limit of traditional data envelopment analysis (DEA) is recognized. Because it only reduces empirical error rather than inherent error, it often leads to overfitting and limits the generalization and accuracy in actual applications. However, it helps evaluate efficiency. These limitations point to more effective strategies, such as Deam, to improve efficiency measurement and reduce bias and accuracy [30].

In total, AI and ML are expected to have a significant impact on various industries. The accuracy limit compared to the person is part of the challenge in recommendation systems. For example, data quality, preprocessing, and the "black-box" characteristics are essential. Big data is heterogeneous, prone to classification errors, and computationally complex. Even in the DEA, the problem of overfitting

and generalization errors remains. These factors provide a more precise understanding and support the continuous development of AI and ML by enabling thorough adjustments to the field's technical and structural aspects.

1.3 Objective of the Study

The general purpose of this study is to design, develop, and implement a Community Extension Service (CES) workflow system with an integration of AI analysis and recommendation for the Saint Michael College of Caraga (SMCC). This System aims to automatically document, approve, and monitor CES tasks to enhance efficiency, increase transparency, and enable data-driven decision-making. Through this System, the report submission process will be streamlined, document management will be more organized, and artificial intelligence will be used to assess prior reports and develop significant project recommendations for CES's three-year development plan.

Specifically, it aims to:

- Design a role-based login system that provides administrator, coordinators and, field survey data encoder separately, and in accordance with their duty accounts.
- Create a reporting module and reporting management that allows coordinators to submit reports, administrators to review, approve or reject them and manage reports by viewing, printing, downloading.
- Include AI recommendation features that generate intelligent recommendations.
- Evaluate the system using ISO 25010 Standards to ensure its functionality, efficiency, and usability.

1.4 Scope and Limitation

The "SMCC Community Extension Service (CES) Workflow System with AI Analysis and Recommendation" project focuses on enhancing and streamlining the report process and approval for the Community Extension Service at Saint Michael College of Caraga (SMCC). In this System, the coordinator may submit and edit their reports, while the administrator is responsible for reviewing, approving, or rejecting them. All approved reports are automatically stored in the database for organization, easy access, and use as the basis for subsequent projects. The System also includes AI analysis and recommendations, in which AI reviews previous reports and data from the Community Needs Assessment (CNA) to suggest projects or programs for inclusion in the CES three-year development plan. The System is web-based, so it can be accessed through desktops, laptops, and smartphones as long as there is a good internet connection. It includes major components such as role-based login (Admin and coordinator), report management, a notification system, and an AI recommendation module to facilitate and accelerate the report production and approval process.

This System is limited to use by the SMCC Community Extension Service and cannot be used by any other department or external organization. AI analysis and recommendations depend solely on the data contained in the System, so the quality of its suggestions aligns with the reported data and the CNA survey. AI is not automatically decided; the administrator's verification and approval are still required before any report or recommendation is finalized. The System requires a stable Internet connection to work correctly, and although it can be used on a smartphone, it is better to use it on a desktop or laptop. Community partners or external users have no direct access to the System; Their data is placed only by coordinators. In future versions, system capabilities such as community access, offline use, and more advanced AI features may expand.



1.5 Definition of Terms

Administrator - A user with full privileges to the System to review, approve, or reject CES reports, manage the coordinator accounts, and view the overall submission history. In the CES Workflow System, only the CES Head or catalogued staff for oversight purposes can serve as the administrator.

AI Analysis - Use of artificial intelligence algorithms to review reports for many aspects, for example, document accuracy, completeness, and consistency. In the CES Workflow System, AI analysis helps coordinators identify missing or incorrect information and provides suggestions for improvement before submission.

AI Suggestions - Refers to automated suggestions for improvement generated by artificial intelligence that analyze input data, identify patterns, and apply either predefined rules and/or learned models. In the SMCC Community Extension Service (CES) Workflow System, AI Suggestions provide coordinators with intelligent prompts to fill in any missing information, improve report readability, and ensure the reports are CES-compliant prior to submission.

Algorithm - A step-by-step procedure or a set of rules in order to solve a problem using a computer. Algorithms check reports using the AI component and provide study recommendations.

Archiving - The long-term storage of data and documents for future reference and record-keeping. Archiving is used inside the CES Workflow System. It ensures that past reports remain available for audits or evaluations.

Artificial Intelligence (AI) –Technology that enables machines to mimic human intelligence to learn, solve problems, and make decisions. The CES Workflow System uses AI to deliver automated feedback and suggest enhancements to coordinators prior to submission.

Barangay. The smallest local government unit in the Philippines. Several CES activities cited in this study are implemented at the barangay level.

Centralized System - A computing architecture in which all of the processing, storage, and management occur at one central location or server, allowing for user control and consistency in access. In the SMCC Community Extension Service (CES) Workflow System, the centralized location means that all CES reports, users, and activity history are stored in a single secure database, thereby simplifying monitoring, maintenance, and updates to the System.

Community Extension Service (CES) – Programs offered by higher education institutions (HEIs) that allow faculty, staff, and students to take their knowledge and skills outside the classroom to improve the quality of life in communities. In the CES Workflow System, reports of these activities are electronically submitted, monitored, and retained in digital format for proper documentation and evaluation.

Community Extension Services (CES) - A systematic process of collecting and analyzing information to identify the most important needs and priorities of a community. The unit of Saint Michael College of Caraga that plans and implements community programs (e.g., trainings, medical drives, tree planting) and documents their outcomes.

Community Needs Assessment (CNA) - An organized digital storage system where data, files, and records are kept for structured access and retrieval. A data-gathering process used by CES to identify

community priorities; in this System, CNA data are checked by AI for accuracy and used to inform recommendations.

Coordinator - Administrator or staff assigned the responsibilities of preparing and submitting the CES reports in their department. In the CES Workflow System, Coordinators use predefined templates to prepare reports and submit them to admins for review.

Dashboard - A user interface that visually displays important information and insights. In the CES Workflow System, the dashboard gives administrator immediate access to pending reports, trends, and summaries.

Data Collection - The procedure of acquiring data to make decisions. In CES, it involves acquiring feedback, project results, and report information.

Data Security - refers to the practice of shielding digital information from being compromised, stolen, or downloaded by unauthorized users. In this project, data security ensures that CES coordinators and administrator who report data and generate reports can view or change them only when authorized.

Database - A structured collection of information may lead you to think of other collective databases managed within a computer system. The database in question contains user accounts, reports generated from campers, and historical records located in the CES.

Database/Repository - An organized digital storage system where data, files, and records are kept for structured access and retrieval. The structured digital storage where all reports, CNA data, and activity records are saved, organized, and retrieved.

Digitalization/Digitized Reporting - The process of converting paper-based or manual records into digital formats to enable faster, more accurate, and more efficient handling. The CES Workflow System replaces traditional paper submissions with digitized reporting. Coordinators submit reports online, making the process faster, less error-prone, and easier to track, while administrator can review, approve, and archive reports more effectively.

Electronic Document Management System (EDMS) - A type of software that electronically captures, stores, organizes, and retrieves documents for better access and control. The proposed CES Workflow System adopts principles from EDMS by allowing digital storage, indexing, and retrieval of reports. This ensures that CES documents are accessible, well-managed, and preserved for institutional and community use.

ISO/IEC 25010. An international software quality model was used in this study to guide testing of the System's functionality, reliability, performance efficiency, usability, security, maintainability, and portability. In this study, ISO/IEC 25010 serves as the benchmark for testing and evaluating the CES Workflow System. It ensures the System is not only functional but also efficient, secure, user-friendly, and sustainable in the long run.

Local Area Network (LAN) – A network connecting devices in a given area. Within the CES Workflow System, it enables the System to operate on SMCC grounds without using the internet, provided it is located within the institution's local network.

Machine Learning (ML). A subset of AI that learns patterns from data; conceptually related to the System's AI functions but constrained here to analysis and recommendations (no automatic decisions). In the CES Workflow System, ML concepts are applied in a limited way: AI analyzes CNA data for

accuracy and generates project recommendations. The System does not make automatic decisions; instead, it provides insights to assist administrator in planning.

Maintenance - refers to the act of repeatedly updating and repairing a system to sustain its performance and stability. For this project, maintenance will be defined as the ongoing process of ensuring the CES Workflow System runs and remains properly maintained and secure over the long term.

Paper-Based System – represents a conventional document management approach that uses physical paper, yet produces delays and errors, and exposes data to loss risks. The CES Workflow System replaces the outdated process with a centralized digital platform that enhances efficiency and accuracy and provides better accessibility.

Report Status - The state of a submitted report in the workflow: Pending (awaiting review), Approved (meets standards), or Rejected (requires revision/resubmission). The CES administrator can check the Pending reports for the coordinators' submitted forms that can be approved by the administrator or rejected, and write a note for the rejected form that can be re-edited and resubmitted.

Role-Based Access Control (RBAC) – A security feature that grants system access based on roles. Within the CES Workflow System, RBAC ensures that coordinators and administrator have access only to the features required for their functions.

Three-Year Development Plan (3YDP) - A structured, forward-looking document that outlines strategic goals, objectives, and actionable steps for an individual, organization, or project to achieve significant progress over the next three years. In the CES planning document for community projects, the System's AI provides suggestions that can inform this plan.

User Interface (UI) - A system that allows user input and access to its functions. The System has a visual interactive element too. The UI in this study presents accessible dashboards, forms, and tools to do so. Coordinators and administrator can also utilize.

Workflow System - A digital platform that automates, manages, and tracks processes that define tasks, roles, and a path for data streams. Submitting, approving, and tracking reports are included in the workflow system within the CES context.

CHAPTER 2

METHODOLOGY

2.1 Research Design

This study will use a Developmental Research Design (Systems Development Research)**.** It focuses on designing, developing, and evaluating the SMCC Community Extension Service (CES) Workflow System, with AI Analysis and Recommendation. The research will follow the Design and Development (D&D) method, ensuring the System's functionality and usability.

This study employs a Developmental Research Design (Systems Development Research), which is suitable because it focuses on creating and evaluating an innovative technological solution. This method was employed to design, develop, and review the SMCC Community Extension Service (CES) Workflow System, incorporating AI Analysis and Recommendations. The design is appropriate because the study aims to generate a functional and user-centered system that automates report management and includes artificial intelligence for reviewing and providing recommendations. By following the Design and Development (D&D) method, the researchers can systematically plan, construct, and evaluate the system in iterative phases, ensuring that the final output meets the criteria of functionality, usability, and efficiency based on ISO/IEC 25010 software quality standards.

2. 2 Software Development Life Cycle



Figure 1. Agile Methodology

The System will use the Agile Software Development Life Cycle (SDLC) to ensure incremental, flexible, and user-centered development. Through iterative sprints, each system module—such as report submission, approval process, and AI recommendations—will be designed, ideally developed, reviewed, and refined based on actual feedback from CES coordinators and admin staff. This iterative approach enables quick identification and correction of issues during development. Regular stakeholder meetings will also be held to maintain alignment with institutional requirements, functional goals, and usability standards. In this way, the development cycle accelerates as the quality of each part of the System is certain.

2.3 System Architecture

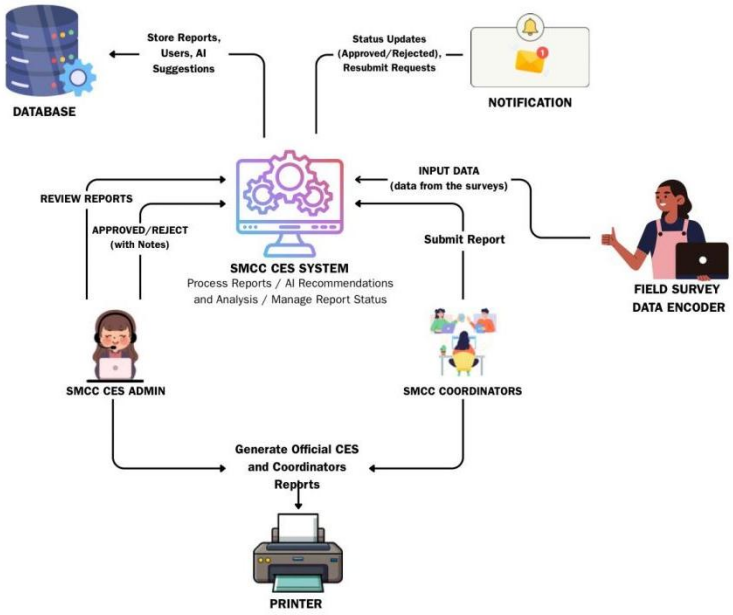


Figure 2. System Architecture

Figure 2 shows the architecture of the proposed system. Users (Field Survey Data Encoder, Coordinator, and Admin) connect to the web-based CES Workflow System through a role-based login. the Field Survey Data Encoder enters and submits survey data into the system, which the Coordinator uses to prepare and submit community extension reports stored in the database. the Admin reviews, approves, or rejects reports, and the system automatically sends notifications. Once approved, the system allows printing and saving the report; if rejected, it displays the Admin notes with the option to resubmit. Additionally, the system has an AI feature that analyzes survey data and completes 3-Year Development Plans to provide intelligent recommendations.

2.4 Conceptual Diagram

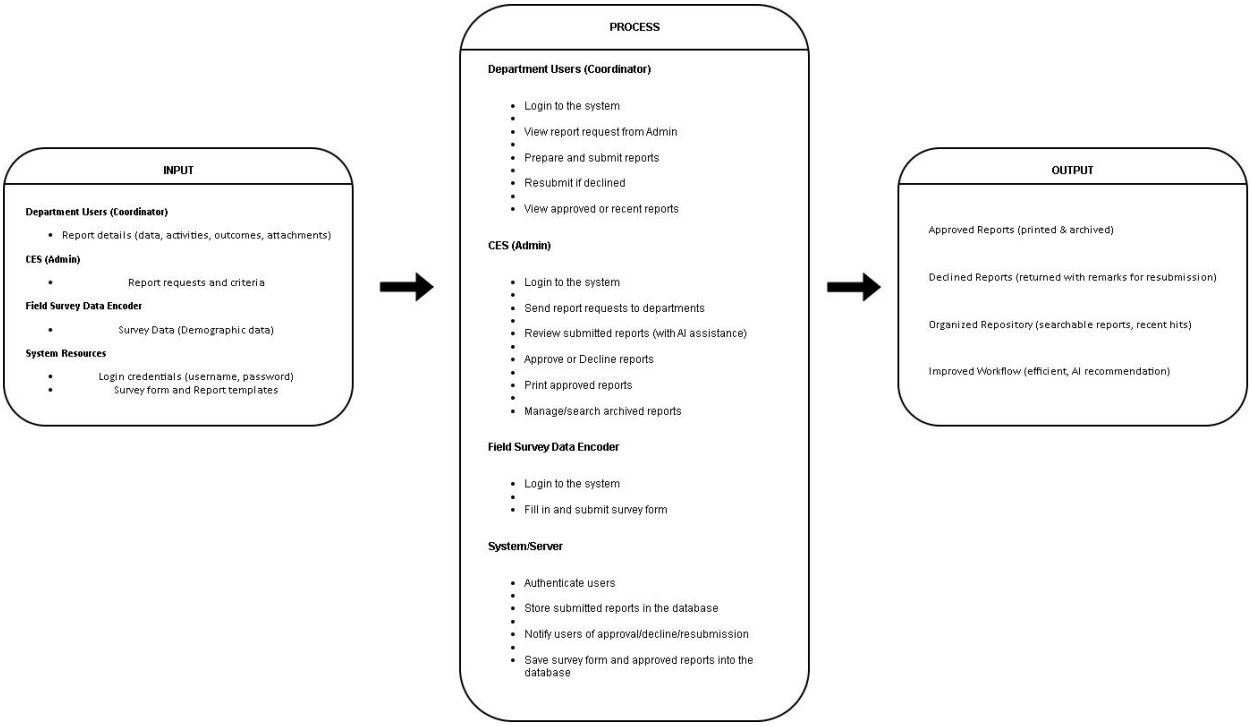


Figure 3. Conceptual Framework

Figure 3 describes the system for managing reports and surveys. There are three user types: the Department Coordinator, who prepares and submits reports; the CES Admin, who requests and analyzes reports and the Field Survey Data Encoder, who enters data from the survey.

The system runs the entire process: authenticating users, storing data, and using AI for outreach program recommendation. The results are approved reports that are archived, rejected reports with feedback, and a more organized and efficient workflow for everyone.

2.5 Algorithms / Framework Used

In today's digital era, it is important to use effective tools and frameworks to build a smooth and effective web-based and AI-driven system. In the SMCC Community Extension Service (CES) Workflow System with AI analysis and recommendation features, frameworks provide structure, pre-built components, and standardized processes that speed, facilitate, and improve development.

Frameworks:

- Bootstrap is an open-source front-end framework used to create responsive and mobile-friendly web pages.
- Sentence Transformer is an advanced model in natural language processing that converts sentences or paragraphs into numerical forms, which represent the overall meaning of the text. Unlike traditional methods based only on words, it understands context and semantics to determine the similarity of sentences even if the words used are different.

Metric

- Cosine Similarity is a way to measure how similar two texts or sentences are based on their meaning. It is used after sentences are converted to numerical embeddings using the Sentence Transformer. In other words, cosine similarity measures the angle between two vectors — the smaller the angle (the closer the result is to 1), the more similar the meaning of the two sentences.

AI training Flow chart:

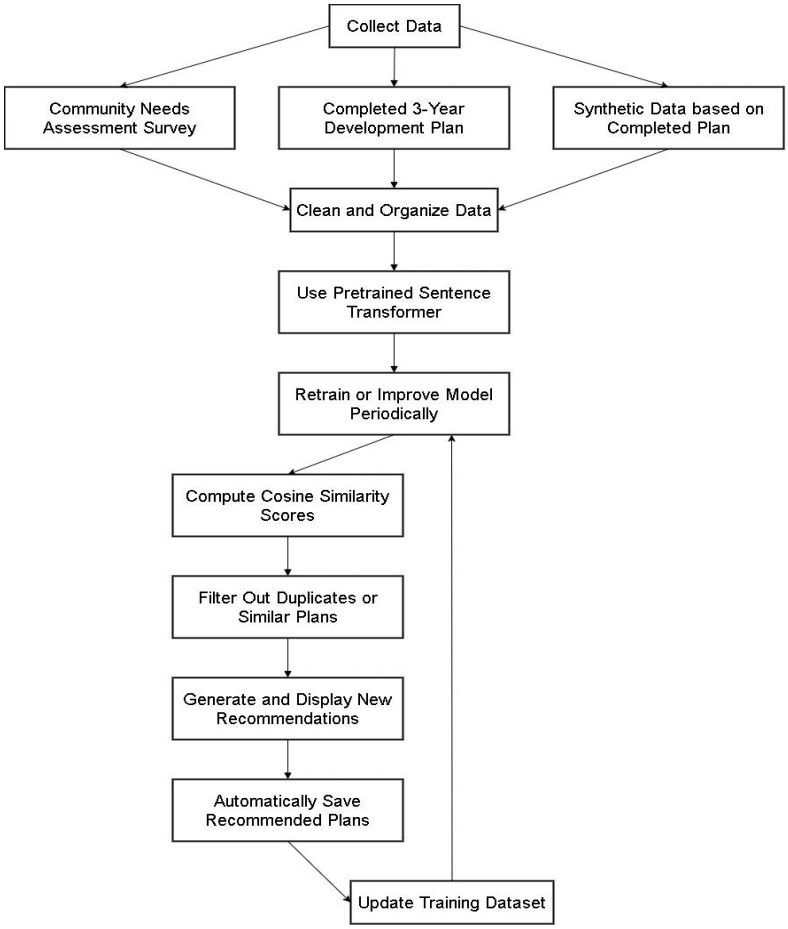


Figure 4 AI training Flowchart

figure 4 shows the training process for the AI recommendation system using data from the Community Needs Assessment Survey, the 3-Year Development Plan, and a synthetic dataset. First, the data is collected and cleaned before combining it into a training dataset. Then, the AI model is trained using cosine similarity to generate recommendations based on community needs. When new data is available, the AI automatically generates and saves a recommendation to continuously improve the model.

2.6 Use Case Diagram

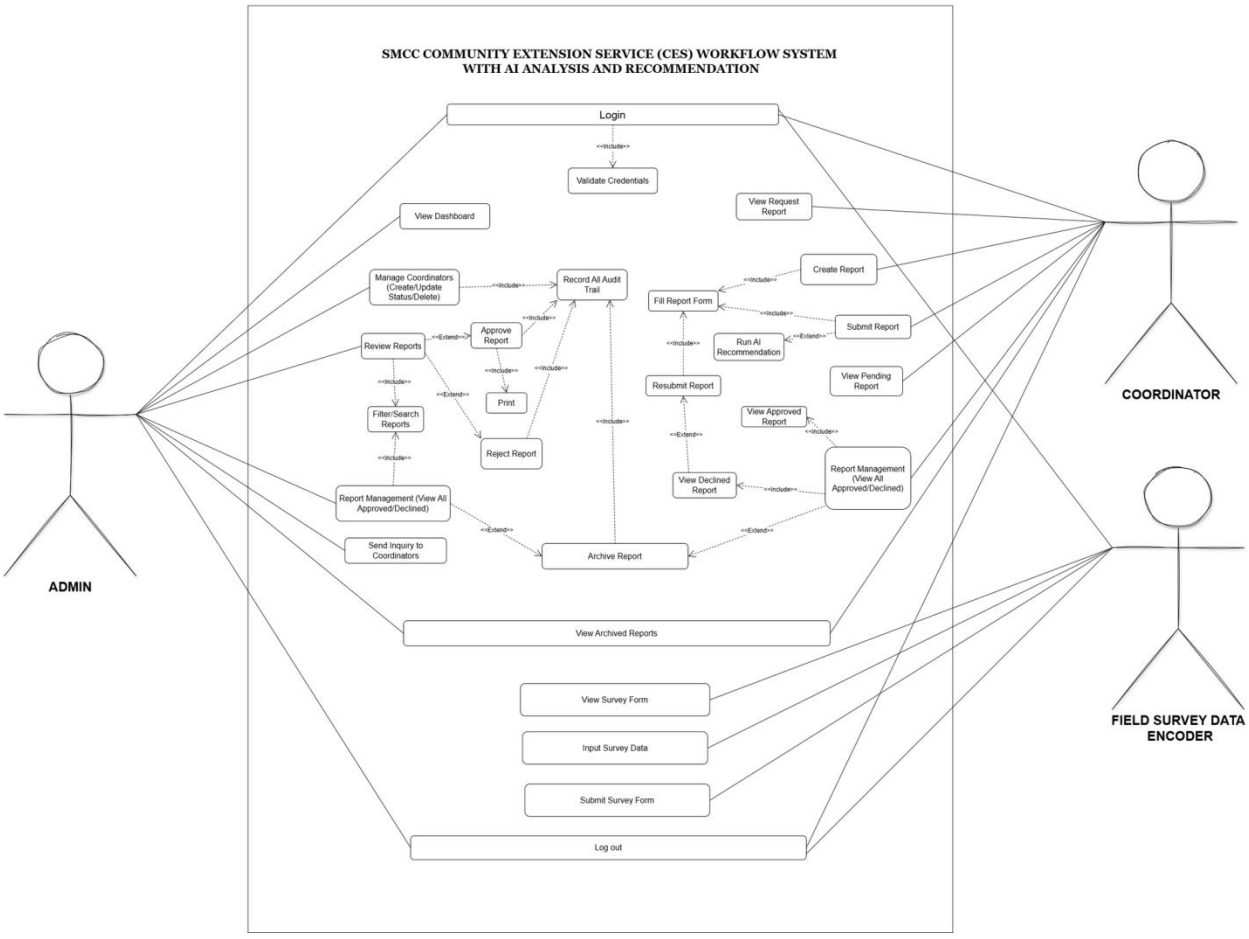


Figure 5. Use case diagram

Figure 5 describes the use-case diagram for the SMCC Community Extension Service (CES) workflow system with AI analysis and recommendation. Here are the three main actors: the admin, the coordinator, and the field survey data encoder, each with their own duties and accessions within the system. These activities are organized to clearly show how they interact with the system features.

The admin`s interaction with the system includes managing coordinators' accounts, viewing the total number of reports (pending, approved, declined), reviewing and approving or rejecting reports, sending inquiries to coordinators, and managing reports, including printing and archiving them.

Meanwhile, the coordinator can make a report, submit a report, and resubmit when it was rejected. The coordinator can also look at requested reports, pending reports, approved reports, and declined reports. It also has the ability to use AI recommendations for recommending information based on previous data gathered.

The Field Survey Data Encoder, on the other hand, views and fills in the survey form and submits it to the database as data that will then be analyzed by the AI and used for AI recommendations.

This diagram shows the clear interaction of the actors in the system, where the admin focuses on the management and verification of reports and accounts, while the coordinator focuses on making and submitting reports, and providing and submitting demographic data is the field survey data encoder`s action. The AI module also strengthens both sides by providing recommendations to make community extension service reporting more efficient and effective.

2.7 Activity Diagram

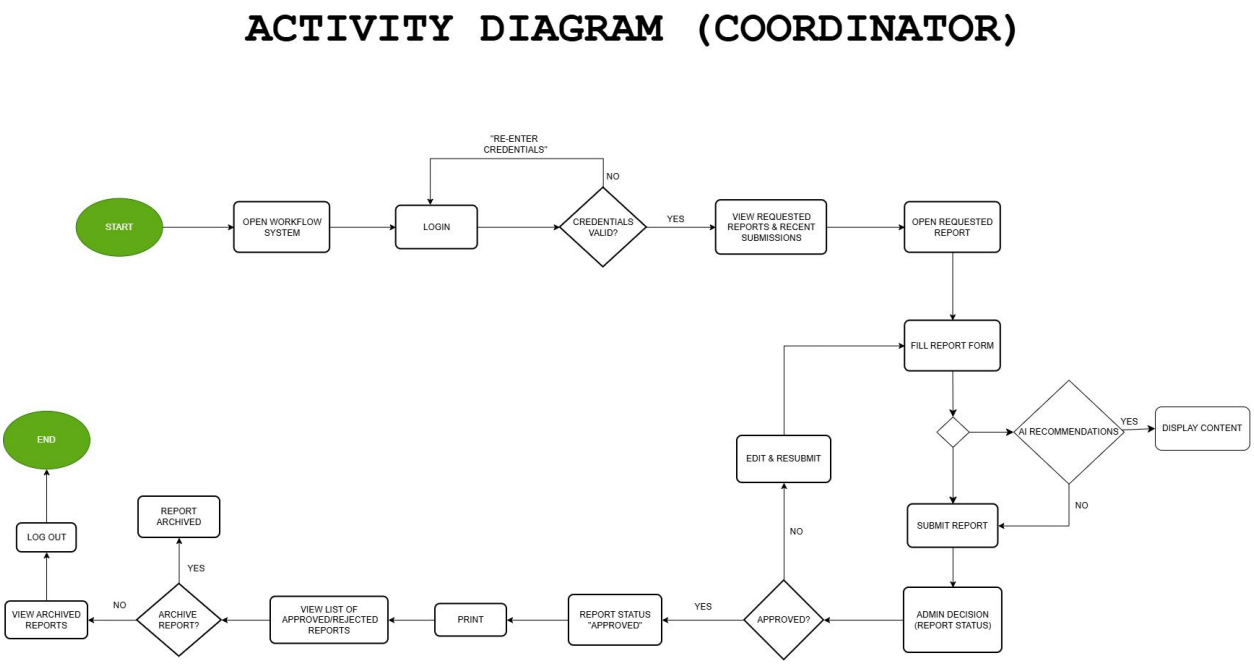


Figure 6 Coordinator Activity Diagram

Figure 6 shows the flow of a coordinator in the use of the workflow system. It starts with the opening of the system and login using the correct credentials. If the information is incorrect, the system will again ask for it to place it correctly once the coordinator is able to see the report requests and previously submitted documents. He can open the required report to initiate the process of developing a new report.

Once the report is selected, filling out the report form is the next step. At this stage, it includes an AI module and provides recommendations. Once complete, the report will be submitted. If not passed, the report may be edited and resubmitted.

In the latter part of the flow, the report will pass the admin decision on whether or not it is approved. When approved, the coordinator can print and save the report as an official copy. There is also an option to archive files as needed. After that, the coordinator can log out, and the process will end.

The diagram shows proper flow from start to finish, which helps to facilitate, organize, and clarify the Community Extension Service (CES) report system.

ACTIVITY DIAGRAM (ADMIN)

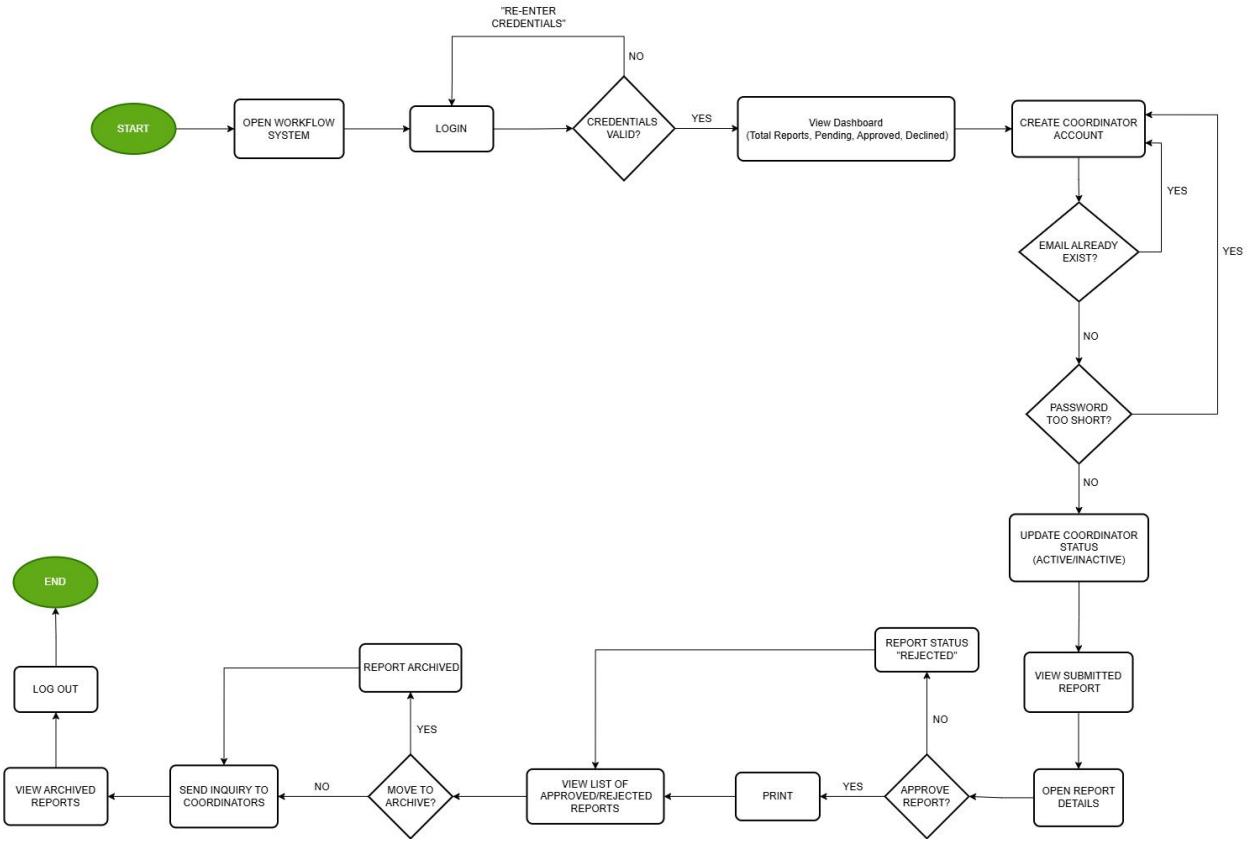


Figure 7 Admin Activity Diagram

Figure 7 describes the activities and duties of the admin within the workflow system. It starts with opening the system and logging in with the correct credentials. If the information is incorrect, the system will again ask for it to place it correctly. Once logged in, the admin will see the dashboard showing the total report, as well as the numbers of pending, approved, and declined reports. From this, he can perform various operations such as managing accounts and processing reports.

One of the admin's main tasks is to create an account for the coordinator. During manufacturing, the system will first check whether the email address already exists or whether the password is too short. Upon passing the standards, the account can be created and updated to reflect its status —active or inactive —ensuring the record management is properly maintained within the system.

In addition to managing the account, the admin also plays an important role in processing the submitted reports. He can look at the details of the report and decide whether it is approved or not. When approved, it can be printed and integrated with the list of official documents. If it does not pass, it can be marked as rejected. The admin also has the option to archive a report and view the list of archived reports. The process ends with logging out, which ensures a safe end of the session.

ACTIVITY DIAGRAM
(FIELD SURVEY DATA ENCODE)

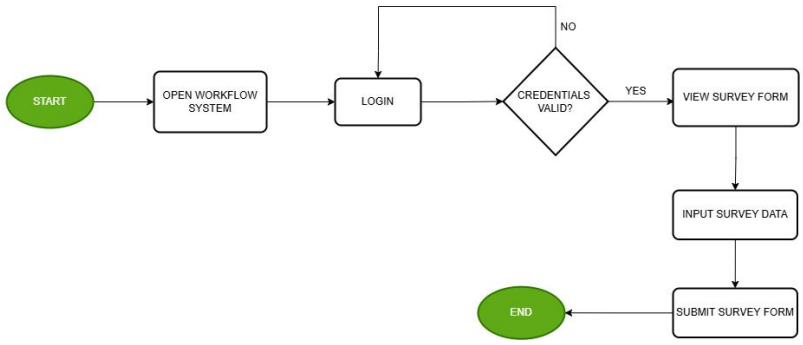


Figure 8 Field Survey Data Encoder (FSDE) Activity Diagram

Figure 8 shows the activity diagram for the Field Survey Data Encoder (FSDE), illustrating the sequence of processes and user interactions within the module. It starts with opening the system and logging in the user. If the credentials are incorrect, he cannot log in; If they are correct, he can access the system.

After successful authentication, the user will be redirected to the digital survey form, the main tool for collecting data from community members. In general, the diagram shows a clear flow from system access to actual field data collection, ensuring an organized and safe process.

2.8 Sequence Diagram

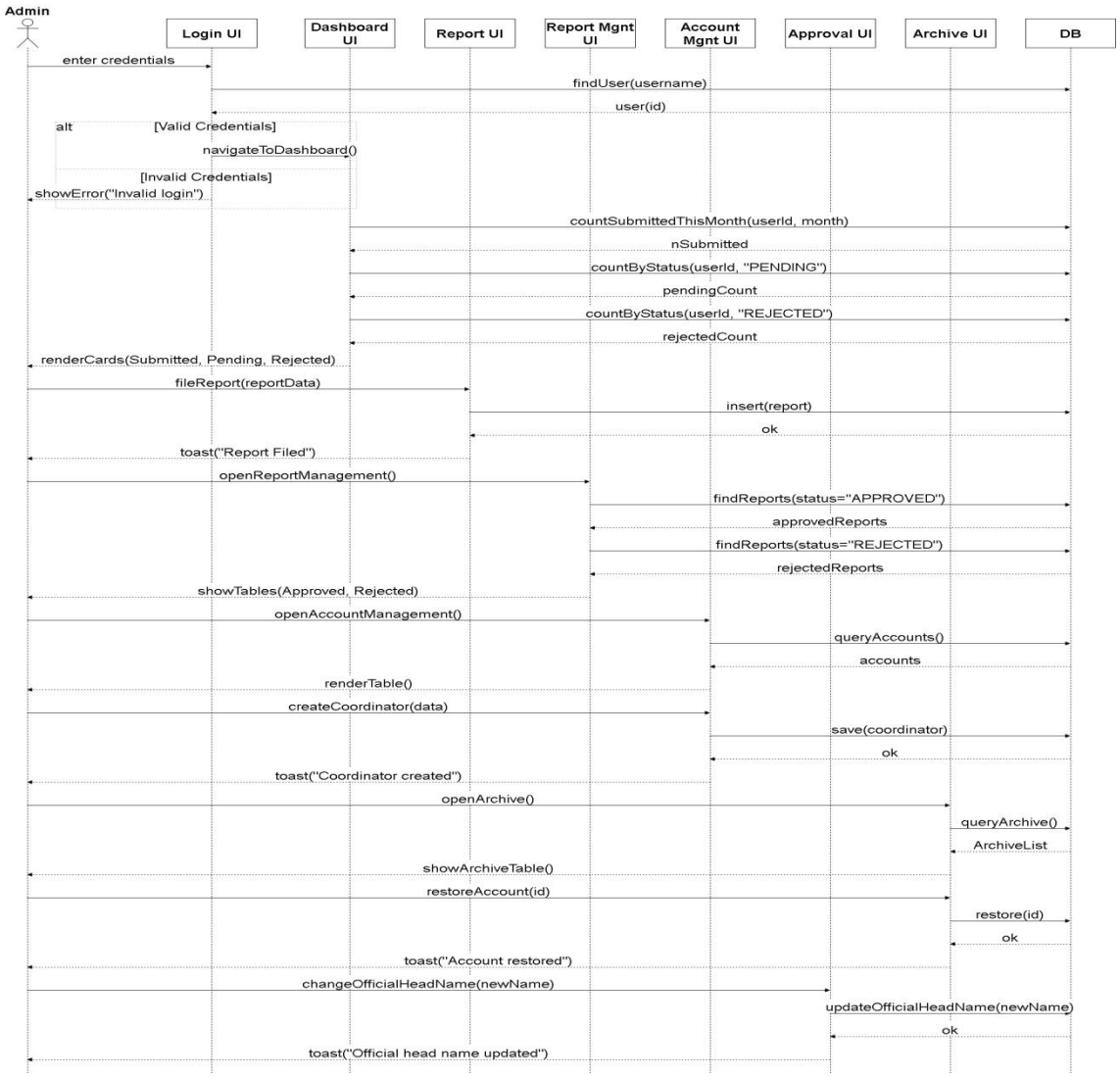


Figure 9 Admin Sequence Diagram

Figure 9 shows a sequence diagram of the administrator's process of using the System. It starts with the user logging in to the Login UI. When the credentials are correct, the user can access the Dashboard, which displays a summary of submitted, pending, and rejected reports retrieved from the Report Service and Repository DB.

The administrator can review reports, manage existing reports, and create a coordinator account using the Account Management UI. Inactive accounts are visible in the Archive UI and can be restored if necessary. In the Approval UI, only the names of the official heads of the report can be changed.

Overall, the diagram shows the administrator's workflow from login → report and account management → approval, → account restoration, ensuring the smooth and effective operation of the System.

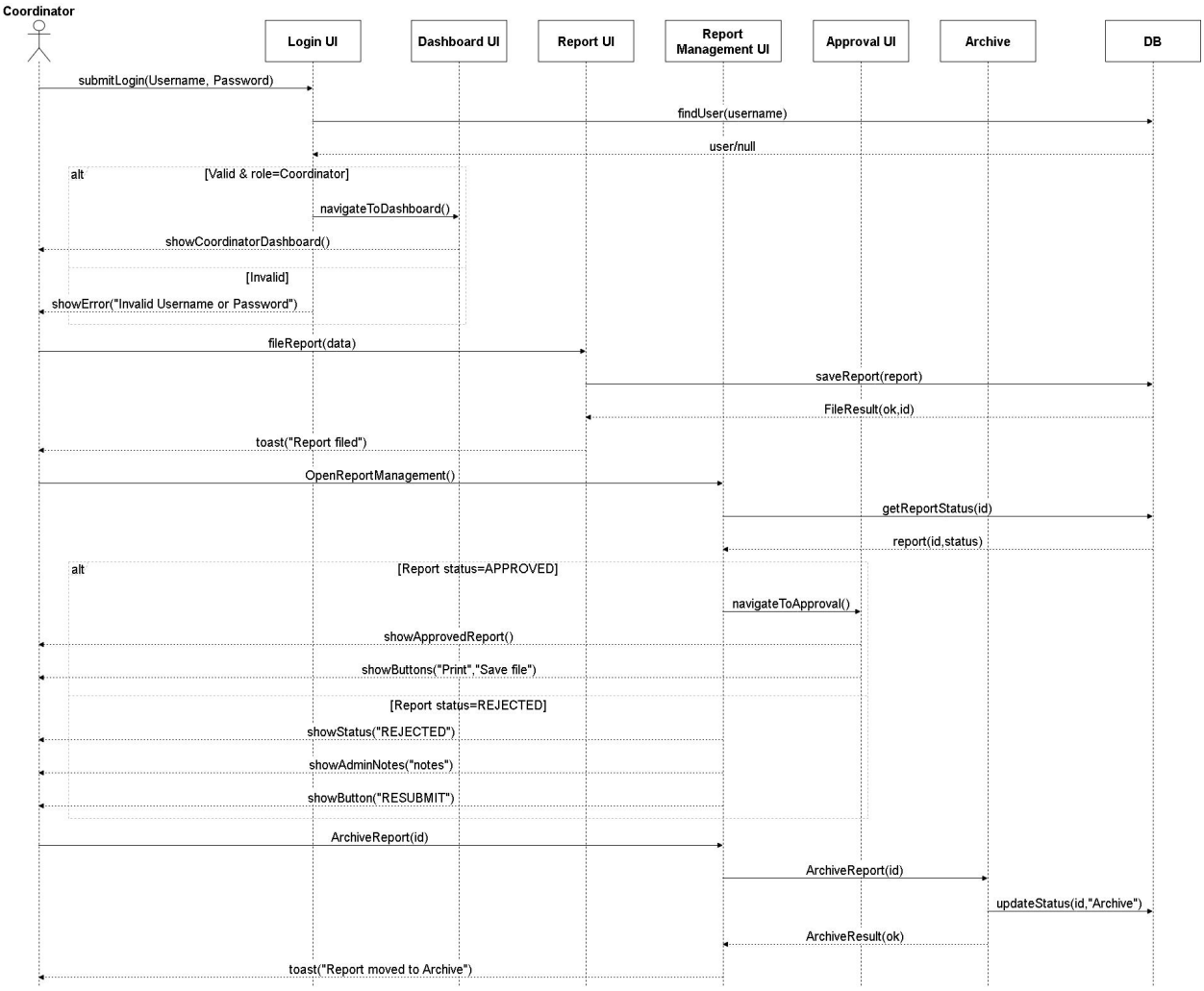


Figure 10 Coordinator Sequence Diagram

Figure 10 shows a sequence diagram for using the System by a Coordinator. The coordinator login process begins using the Login UI. The information goes through the Auth Service for

authentication, and when the user's credentials and role are correct, they can enter the System; if wrong, the message "invalid username or password" will appear.

After login, the coordinator can submit a report in the UI. The data will be sent to the Report Service and saved in the repository/DB as a pending report. Upon successful submission, the coordinator will receive a notification that the report has been filed. They can also open the report management UI to view the status of submitted reports. Once approved, they will see the status and have the option to print or save the report. If rejected, the System will display the administrator's notes and have the option to resubmit the report.

Additionally, the coordinator can archive the report. When removed, the coordinator will also receive confirmation that the report has been transferred to the Archive. In sum, the diagram shows the coordinator's entire flow from login, report submission, status monitoring, resubmission when there is a rejection, to moving the report to the Archive so that the management of reports in the System can be orderly and systematic. systematize the management of reports and user accounts within the workflow system.

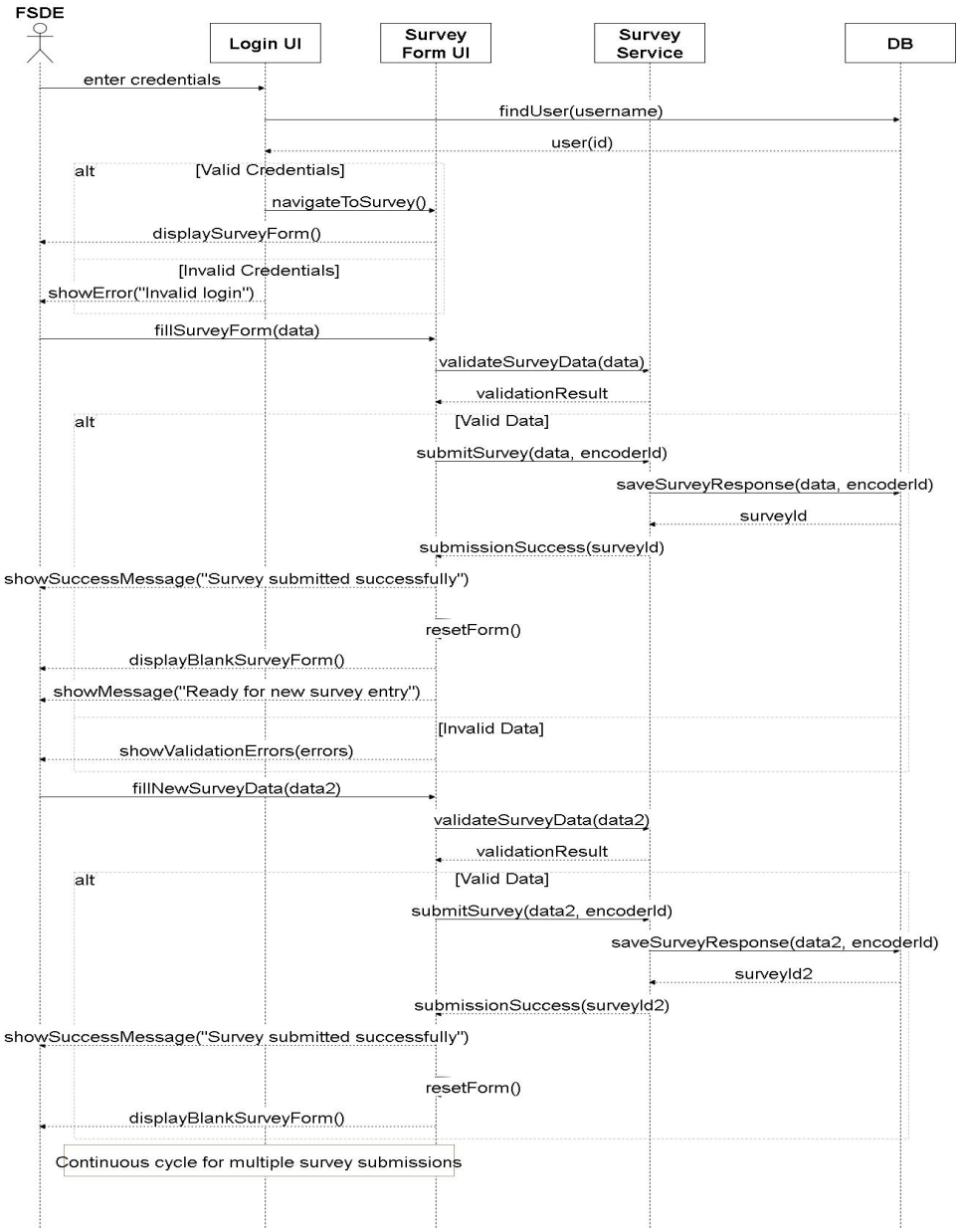


Figure 11 Field Survey Data Encoder (FSDE) Sequence Diagram

Figure 11 shows the encoder logs in to the Login UI, and the DB checks if the credentials are correct. If valid, one can enter the Survey Form UI and see the survey form; if invalid, an error message will appear.

The encoder then fills out the survey and sends it to the Survey Service for validation. If the data is correct, it will be saved to the DB, and a success message will be displayed. The form is reset for a new survey entry, and the process is repeated for each new survey.

In other words, it is a repeated cycle of login → fill survey → validate → submit → reset form for multiple survey submissions.

2.9 ERD (Database Design)

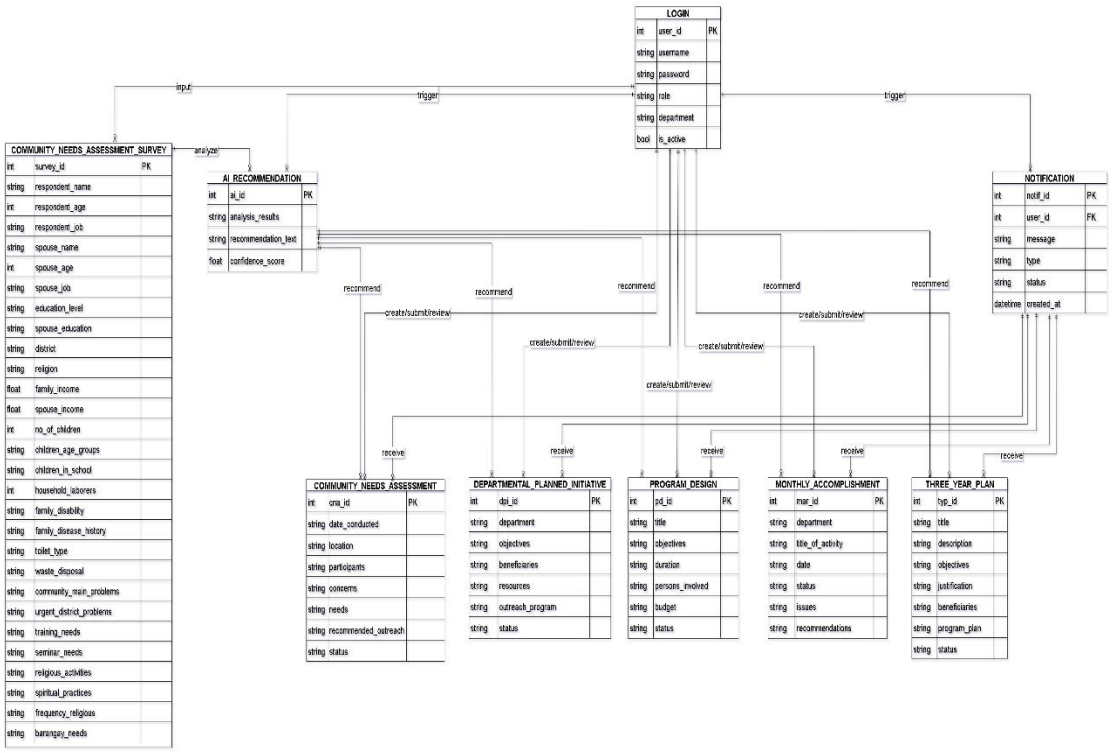


Figure 12 Entity-Relationship Diagram (ERD)

Figure 12 illustrates the Entity-Relationship Diagram (ERD) for the Community Extension Service Workflow System, incorporating AI analysis and recommendations. The diagram contains interrelated entities or tables that support the evaluation, planning, and management reporting processes. The System focuses on the login table, which represents system users, including administrators and coordinators. This includes details such as the username, password, role, and department that determine each user's access.

The primary data source is the Community Needs Assessment Survey. There is important information about respondents, including age, occupation, education, family income, community problems, and other needs. These data are evaluated by the AI Recommendation Module, which provides recommendations based on the review results and previous reports. AI provides suggestions

that can be attributed to various parts of the System, such as Community Needs Assessment, Departmental Planned Initiatives, Program Design, Monthly Accomplishments, and the Three-Year Plan.

The Community Needs Assessment Table contains an official review note from the community, including the date, location, issues, and needs. On the other hand, the Departmental Planned Initiative refers to each department's plans, including goals, beneficiaries, and resources. In the next step, Program Design records a more detailed program design, including the title, purpose, duration, budget, and status. In addition, the monthly accomplishment serves as a record of monthly activities and project reports. At the same time, the three-year plan focuses on a longer-term plan lasting three years and includes justification, beneficiaries, and program plans.

Finally, the notification table serves as the System's mechanism for notifying users about submissions, approvals, and recommendations. Generally, Figure 10 shows that each entity is connected to generate an integrated workflow system. AI utilizes survey data to provide a reasonable recommendation that contributes to more effective planning and conducting of community extension programs.

2.10 User Interface Design (Prototype)

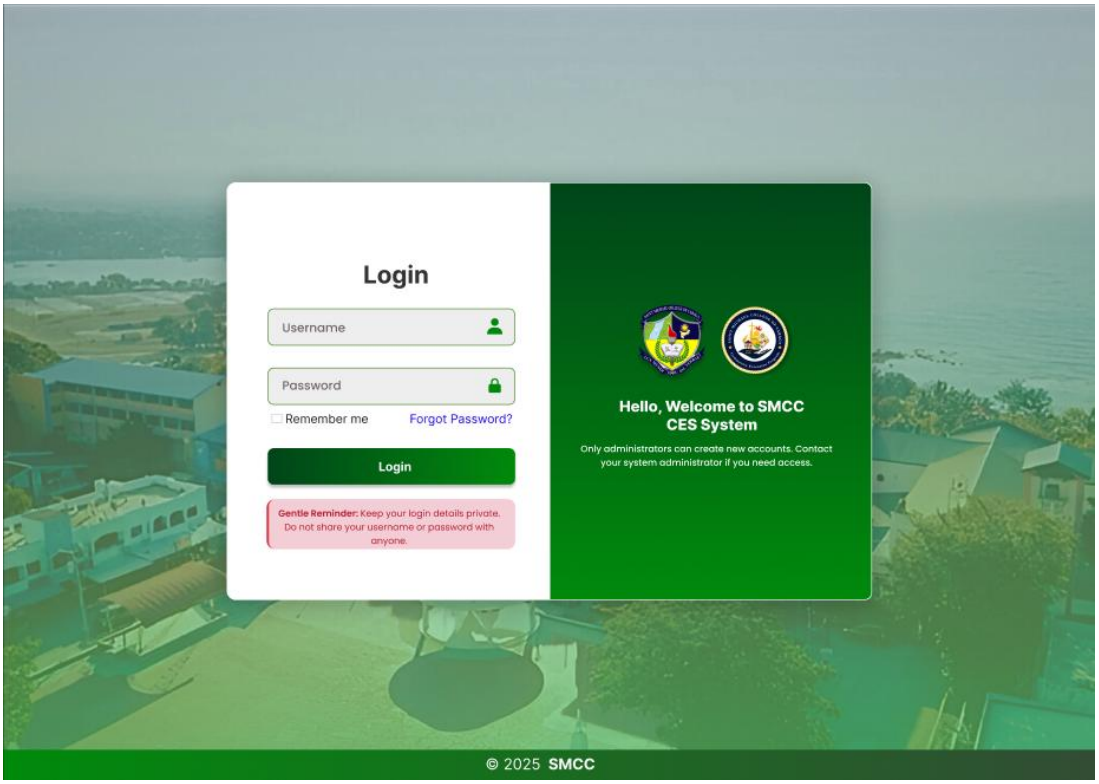


Figure 13 Login Page of the SMCC CES Workflow System with AI Analysis and Recommendation

(Prototype)

Figure 13 shows the login form, which is used for system access. The form has fields for username and password, as well as options like “Remember me” and “Forgot Password?”. There is a clear “Login” button to submit information. The welcome message “Hello, Welcome to SMCC CES System,” and a reminder that those who need access, must contact the system administrator or follow their system code or address. This login form serves as the secure entry point for authorized users of the system.

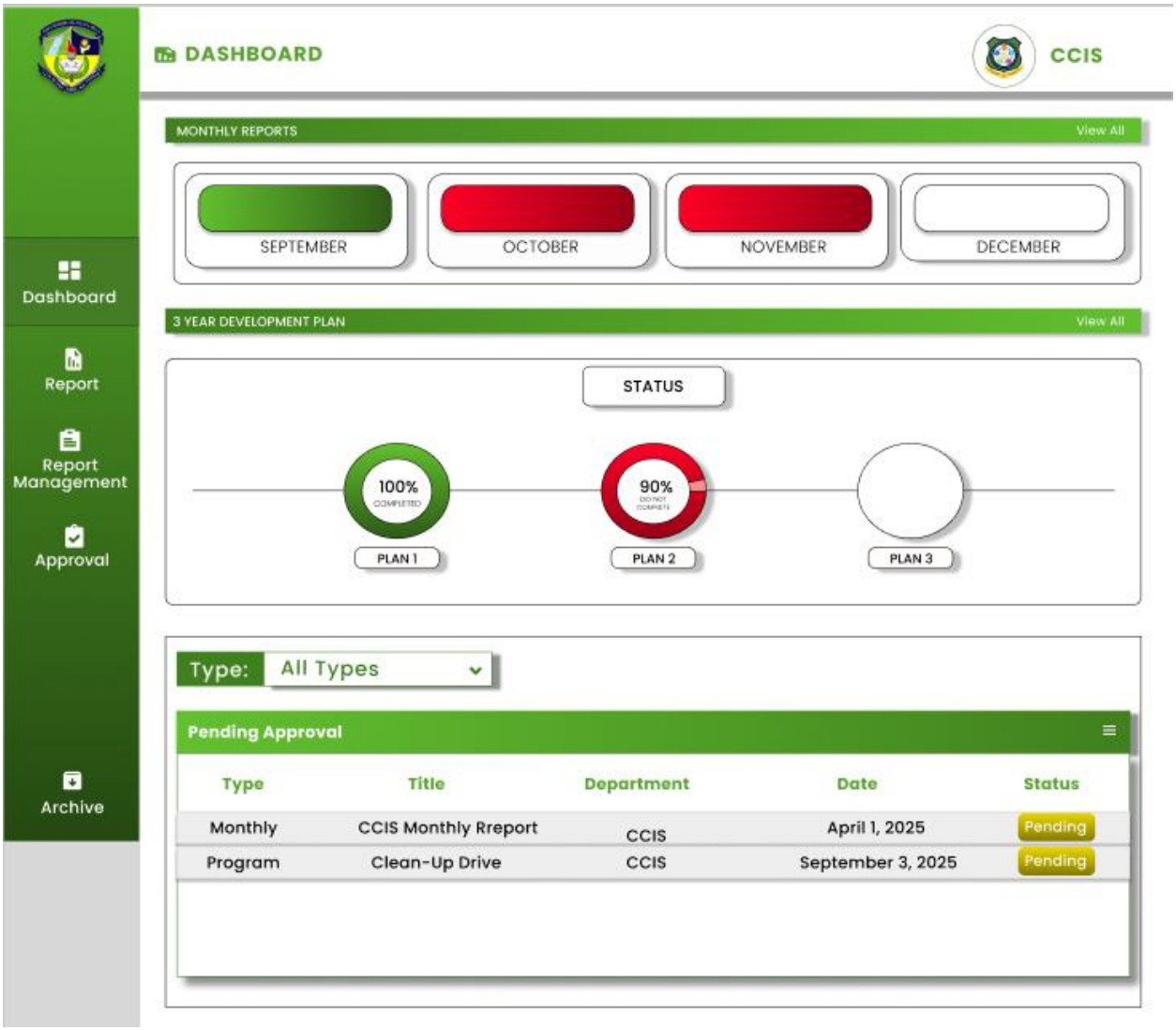


Figure 14 Coordinator Dashboard

Figure 14 shows the dashboard as a centralized monitoring system for tracking institutional reports and development initiatives. It has a monthly reports section that organizes submissions, providing a chronological view of the documentation. The state development plan area shows progress using status indicators and management tools, where ongoing initiatives Plan1, Plan2, and Plan3 can be seen along with their respective completion. In the pending approval section, reports awaiting review are listed.



Figure 15 Coordinator Monthly Report Dashboard

Figure 15 shows a clear overview for the coordinator to easily identify which months the monthly performance report was not submitted. With a list of each month and a simple "submit report" label for submitted months, it is immediately clear if any reports are missing. This will help easily track compliance and identify periods that need to be followed or adjusted.

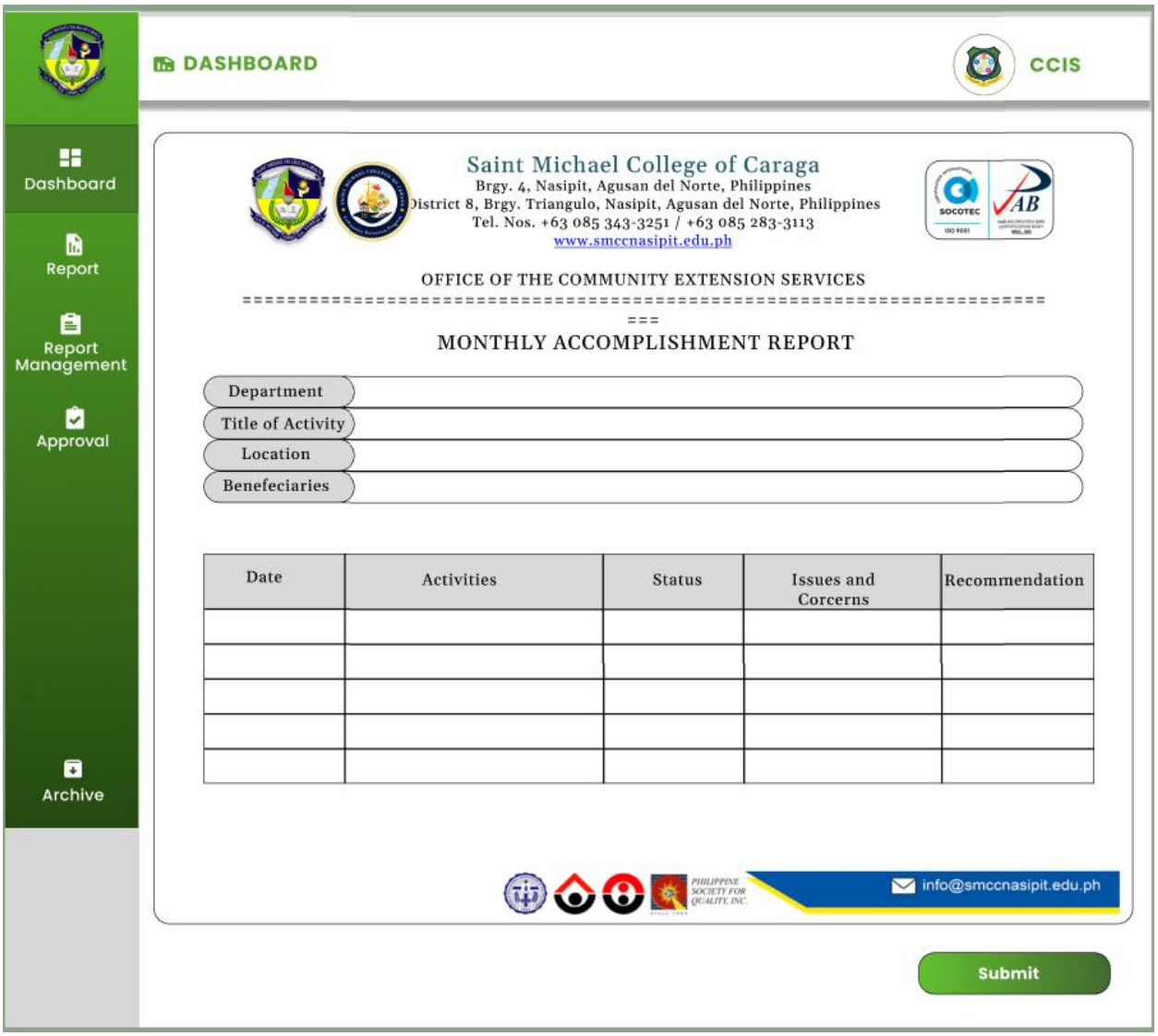


Figure 16 Coordinator Monthly Accomplishment Report

Figure 16 shows the form in which the coordinator makes the Monthly Accomplishment Report. It has a neat template with fields for department, activity title, location, and beneficiaries, followed by a table for logging activities, status, problems encountered, and recommendations. This form ensures consistent, complete reporting of all community extension activities.



Figure 17 Coordinator 3 Year Development Plan Dashboard

Figure 17 shows the coordinator's dashboard for the 3-year development plan, with progress displayed in a legend for easy tracking. It also has a section where the coordinator can post, view, and manage programs and goals to ensure alignment with long-term goals.



Figure 18 Coordinator 3-Year Development Plan Form

Figure 18 shows the form for creating a 3-Year Development Plan. It features an AI recommendation tool that provides intelligent suggestions based on past community surveys and completed plans, to ensure that new strategies are data-driven and based on proven successes.

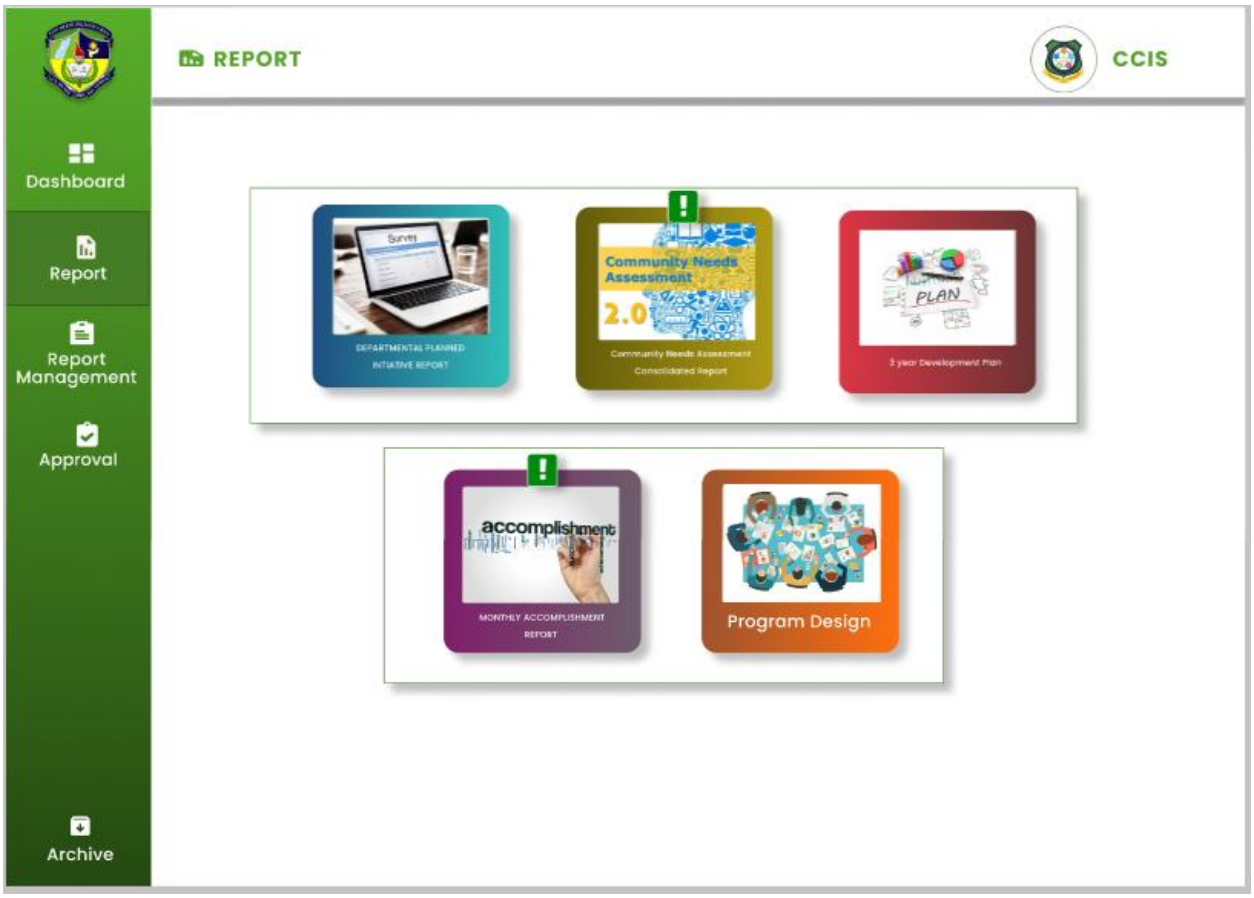


Figure 19 Coordinator Report Navigation

Figure 19 shows the coordinator interface, a comprehensive system for creating and managing reports, designed for coordinators to submit various institutional documents. The interface supports different types of reports, such as Community Needs Assessment with its consolidated version, 3-year development plan, Accomplishment Reports, and Program Design. This streamlined structure allows coordinators to effectively create, organize, and submit reports that directly contribute to both the institution's short-term and long-term development strategies, while maintaining a clear process for review and approval. In addition, there is a notification on the card button that provides immediate information or updates on the status of reports, making monitoring and immediate action on tasks easier.

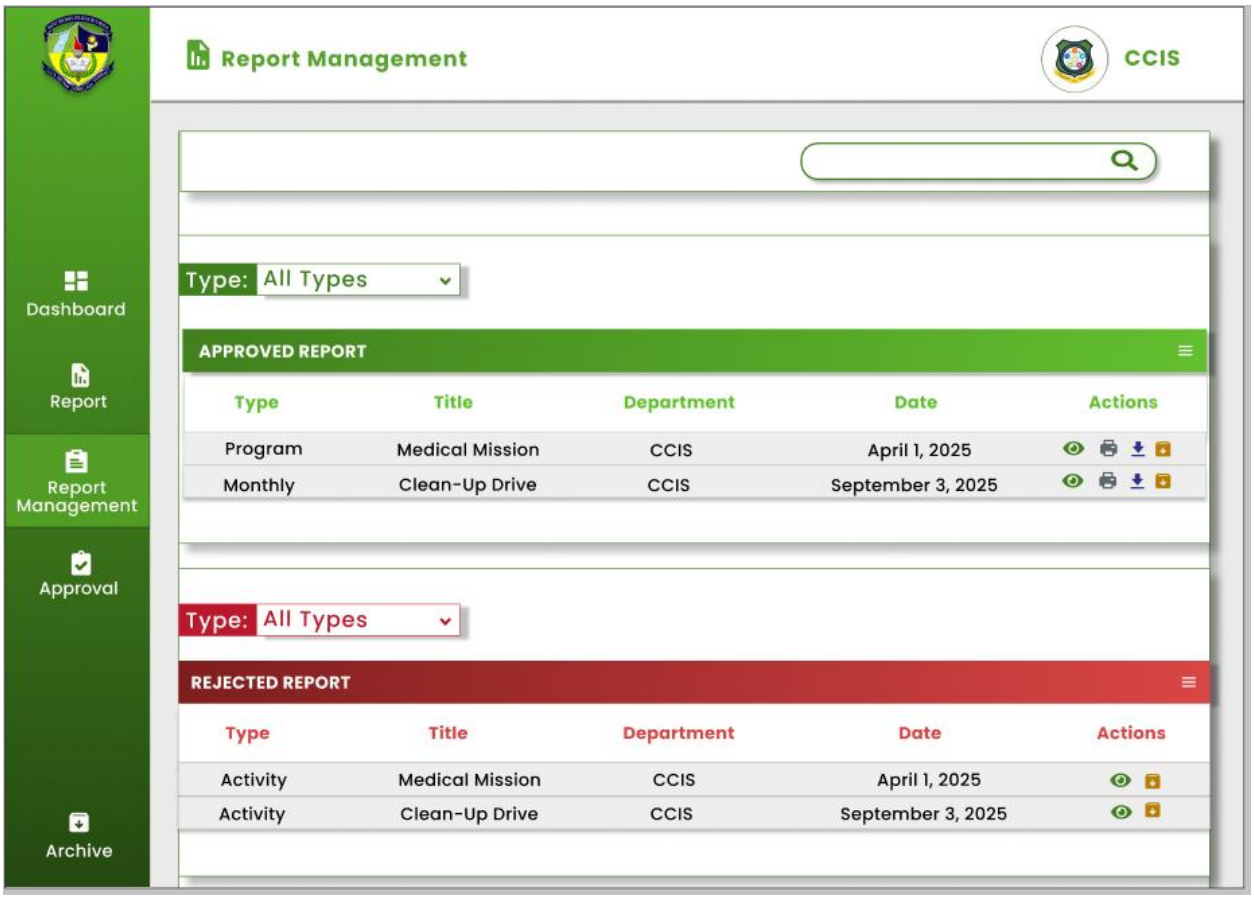


Figure 20 Coordinator Report Management Page

Figure 20 shows the Coordinator's Report Management Page. The interface is divided into two sections: Approved Reports (green) and Rejected Reports (red), arranged in a table with columns for type, title, department, date, and actions.

Each report has action icons such as View (eye), Download (arrow), Print (printer), and Archive (folder) for direct management. There is a filter bar at the top to select a report type and a search bar on the right to find the document quickly. On the left, the navigation panel remains for easy switching to other parts of the system. The design focuses on usability and efficiency, making it easy for coordinators to track and manage their submitted reports.

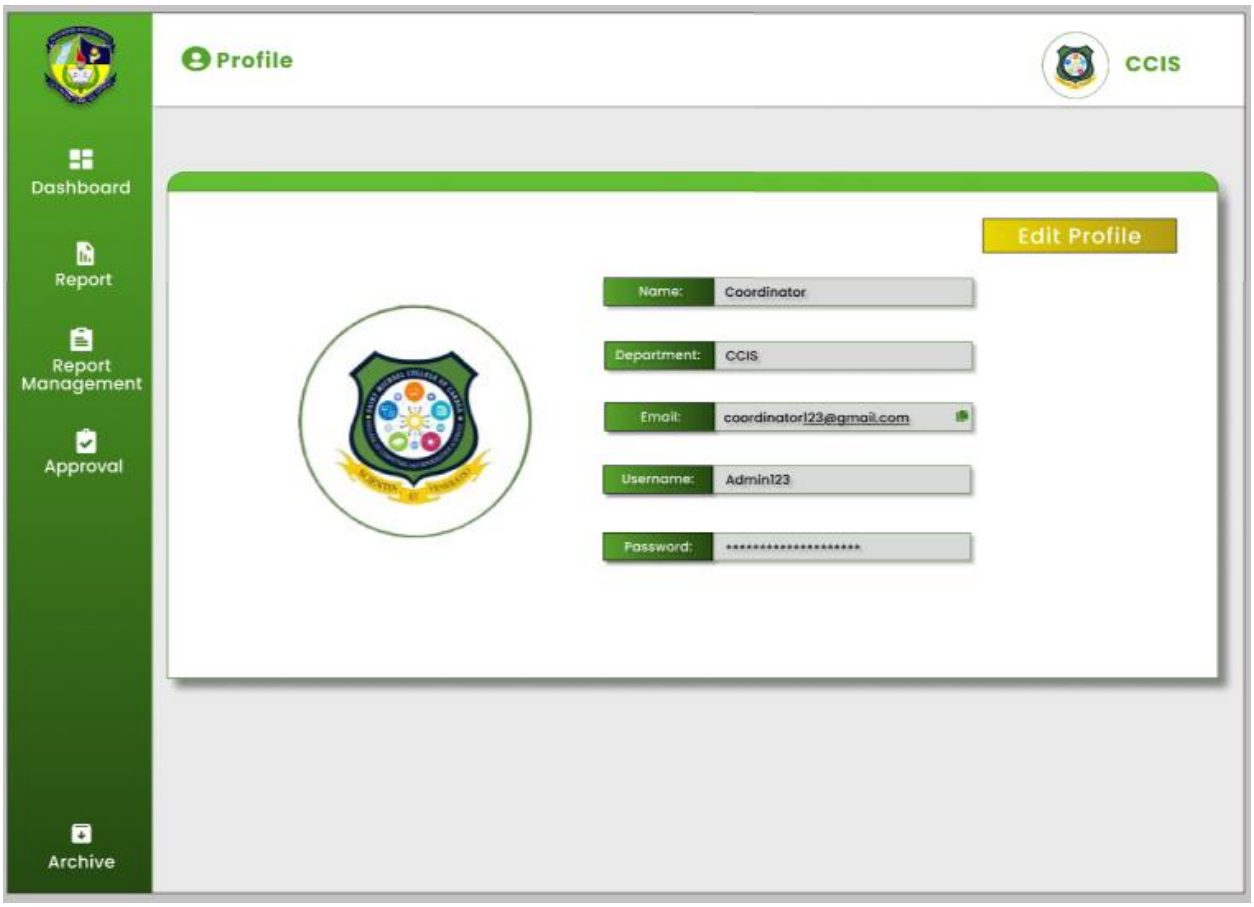


Figure 21 Profile Page (View Mode)

Figure 21 shows the profile interface in view mode, where user information such as name, department, email, username, and password can be found. It has not yet been edited in this Mode. The department logo is on the left. The overall design features a green theme that complements the CES workflow system. The save button is visible but inactive since the data does not change.

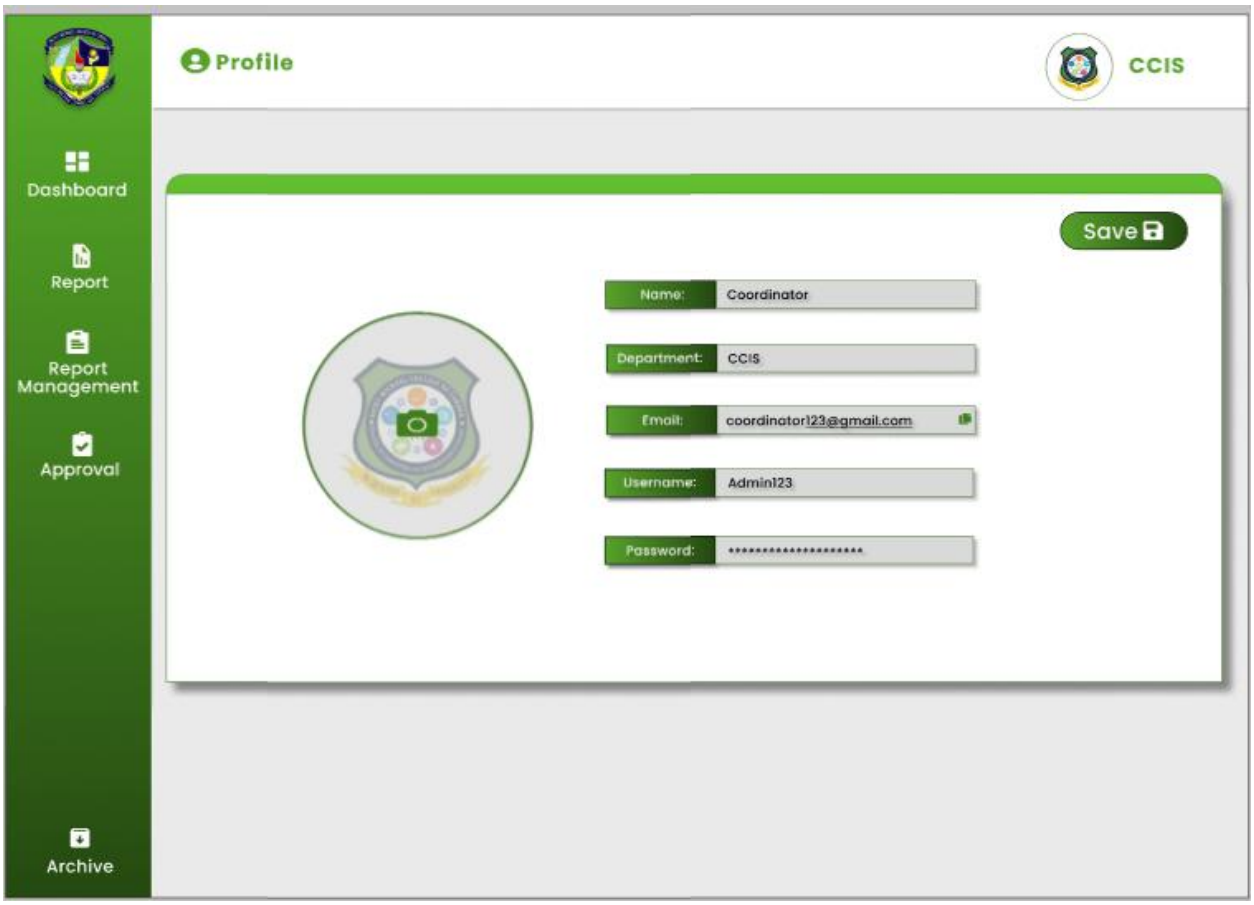


Figure 21-A Profile Page (Edit Mode)

Figure 21-A shows the profile interface in edit mode, which is activated when the edit profile button is clicked. Here, the user can update personal details, including name, department, email address, username, and password. The logo remains on the left and maintains the same layout, making it easy for the user to use and navigate.

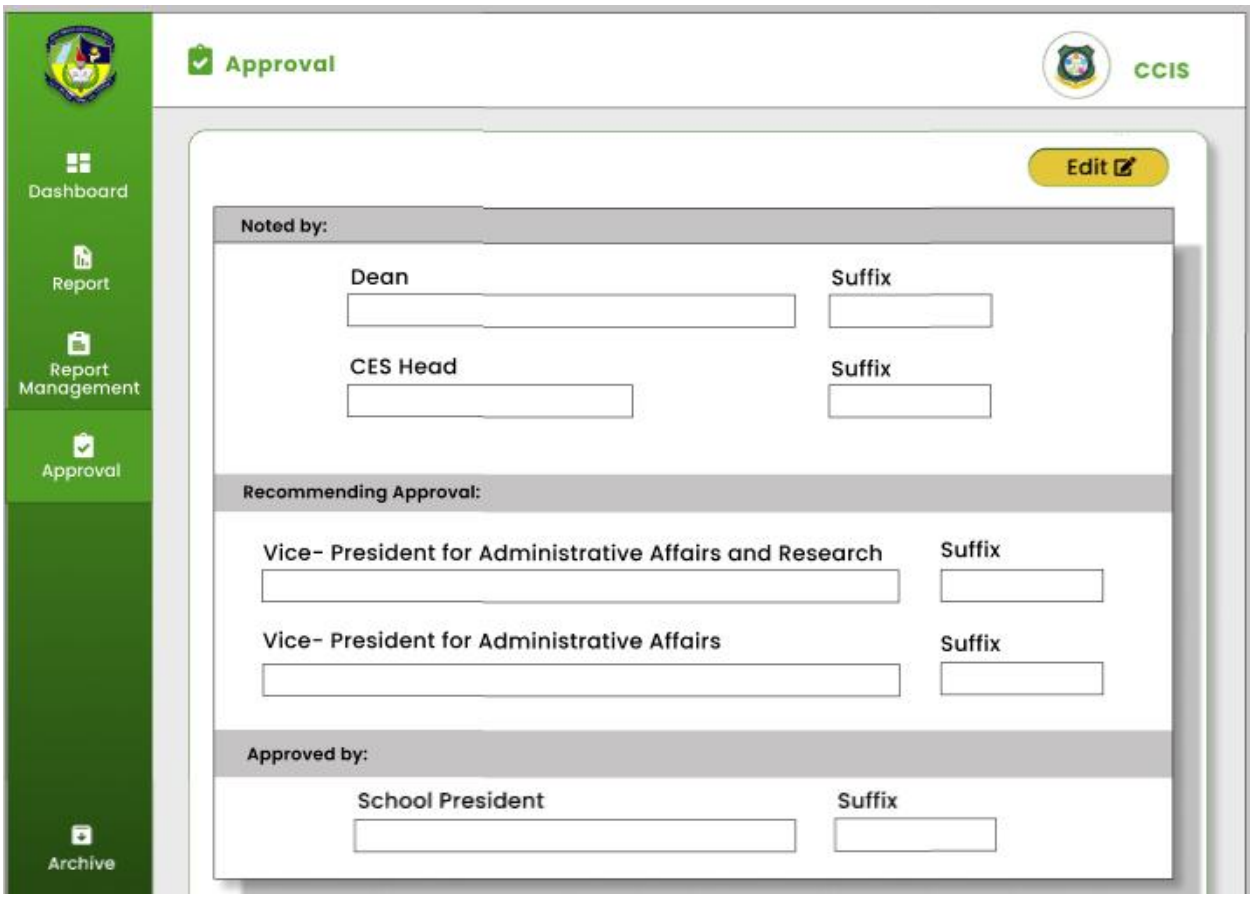


Figure 22 Approval Page (View Mode)

Figure 22 shows part of the approval page reserved for the recommendation and final approval of a project or Report. In this section, the names of the Vice Presidents for Academic Affairs and Research, as well as the Vice President for Administrative Affairs, are reserved for recommendation and approval. This is followed by the School President, who is responsible for providing official approval as the final step of the process. Similar to the rest of the form, this information is automatically saved, appears only in the printed version, and can be edited only when the "Edit" button is pressed.

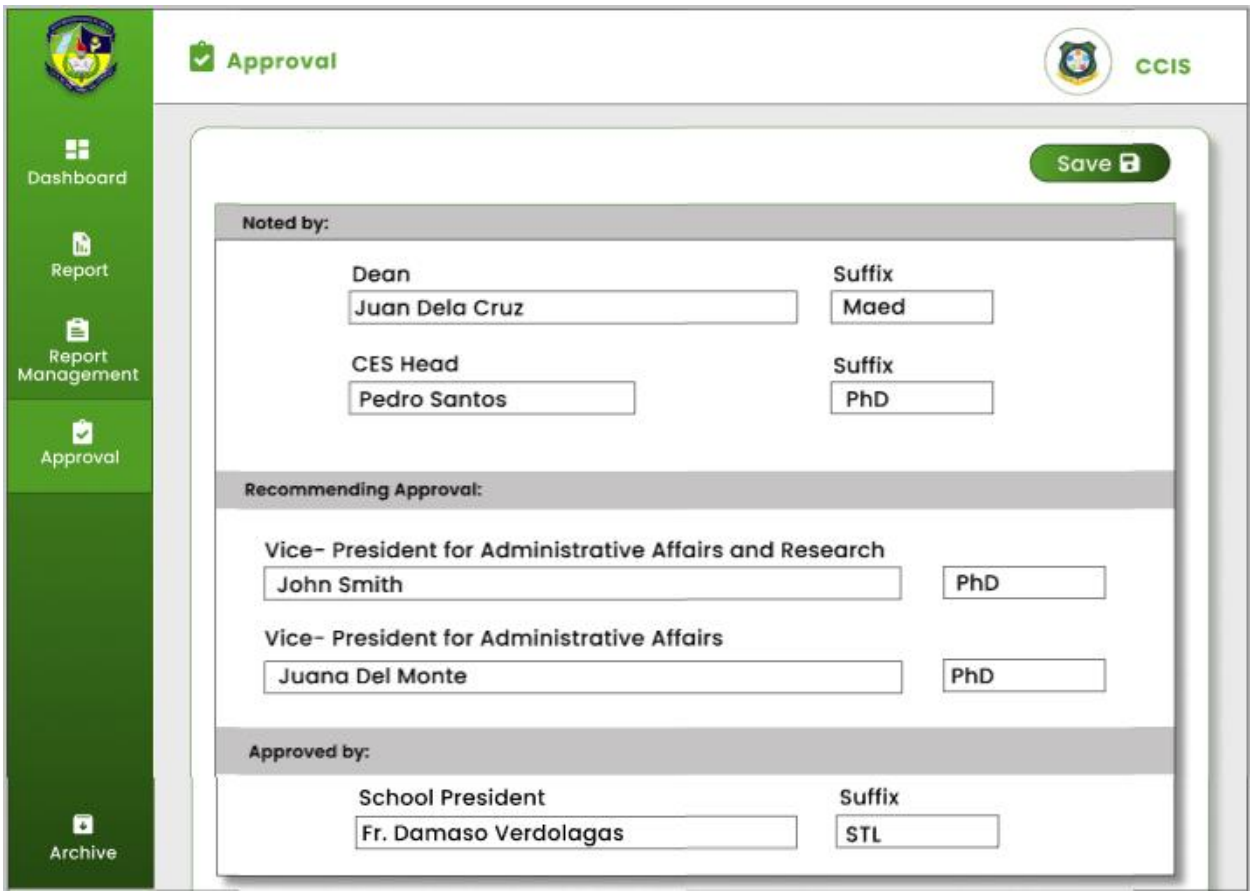


Figure 22-A Approval Page (Filled/Save Mode)

Figure 22-A shows the approval page after the data has been filled in and saved. Here are examples of entries, such as Dean, CES Head, Vice President for Administrative Affairs and Research, Vice President for Administrative Affairs, and School President. The Gray interface shows a straightforward approval process and includes a save button to confirm changes.

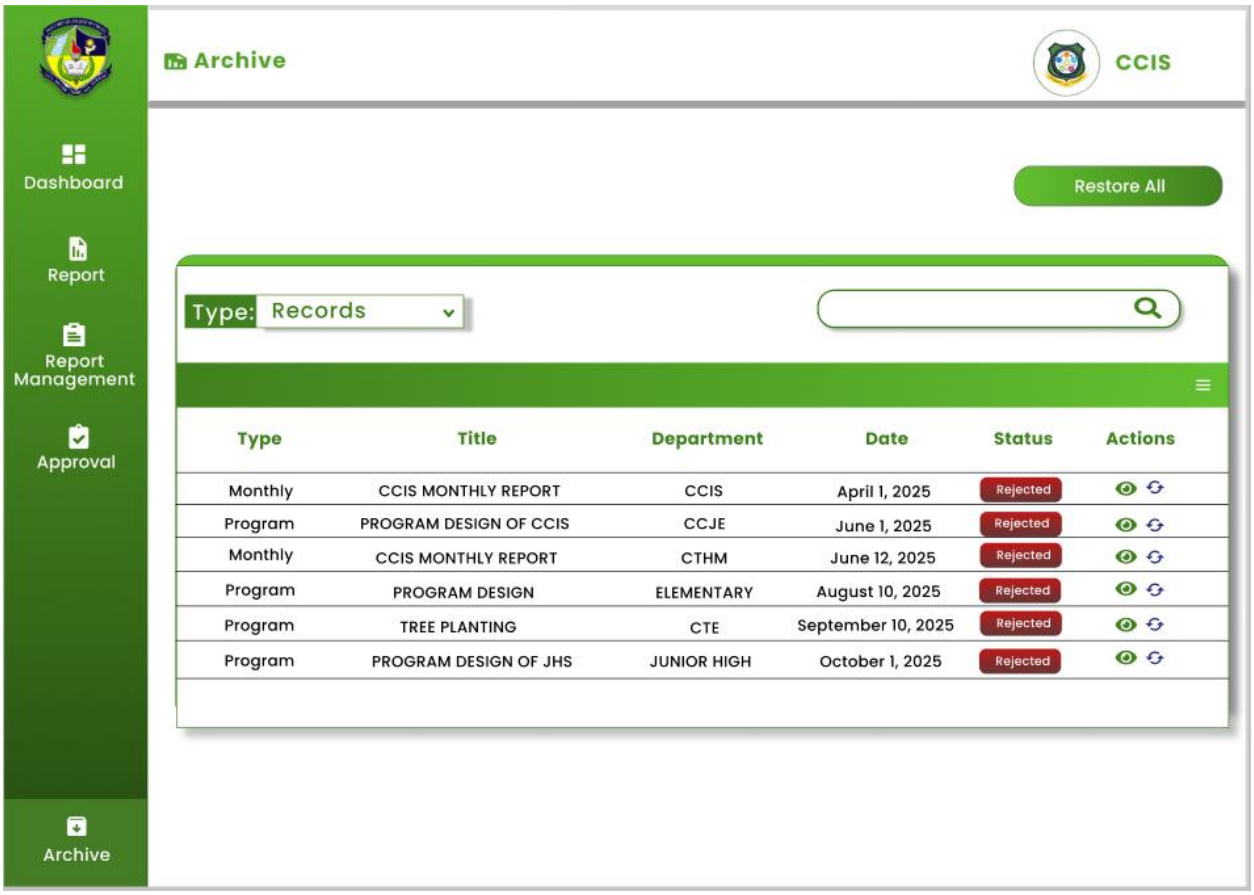


Figure 23 Coordinator Archive Reports Page

In Figure 23, the Archive Page is shown, which serves as a repository for reports transferred by the coordinator. Here, reports can be restored. Above are the “Restore All” button for quick management.

A table with type, title, department, date, status, and actions such as View and Restore is shown in the middle. There are also a filter menu and a search bar for quick searches. This page is important for data recovery.

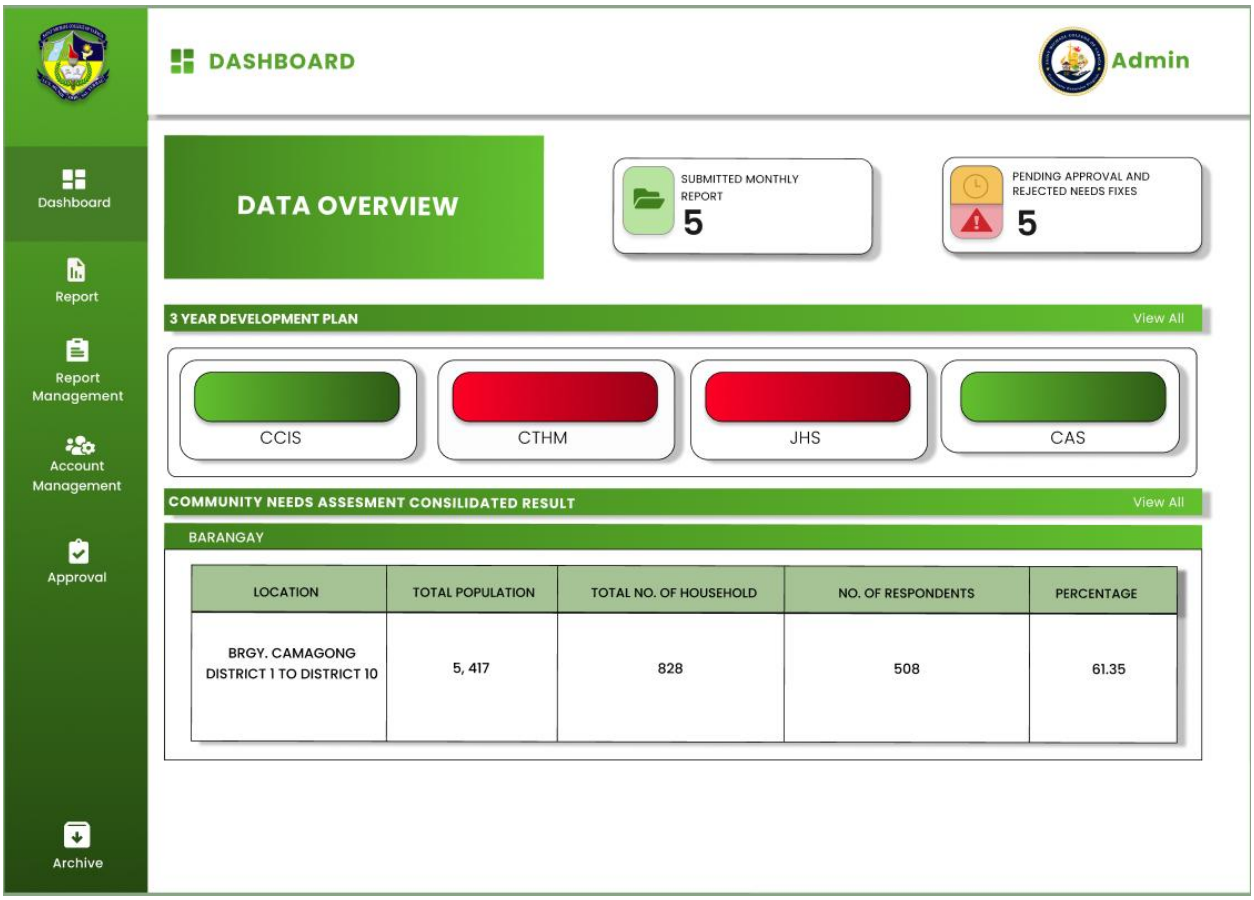


Figure 24 Dashboard (Administrator Interface)

Figure 24 shows the main admin dashboard, which serves as a centralized overview. Here, the admin can track each department's 3-Year Development Plans, view consolidated community survey results, and access management tools for reports and accounts all in one interface.

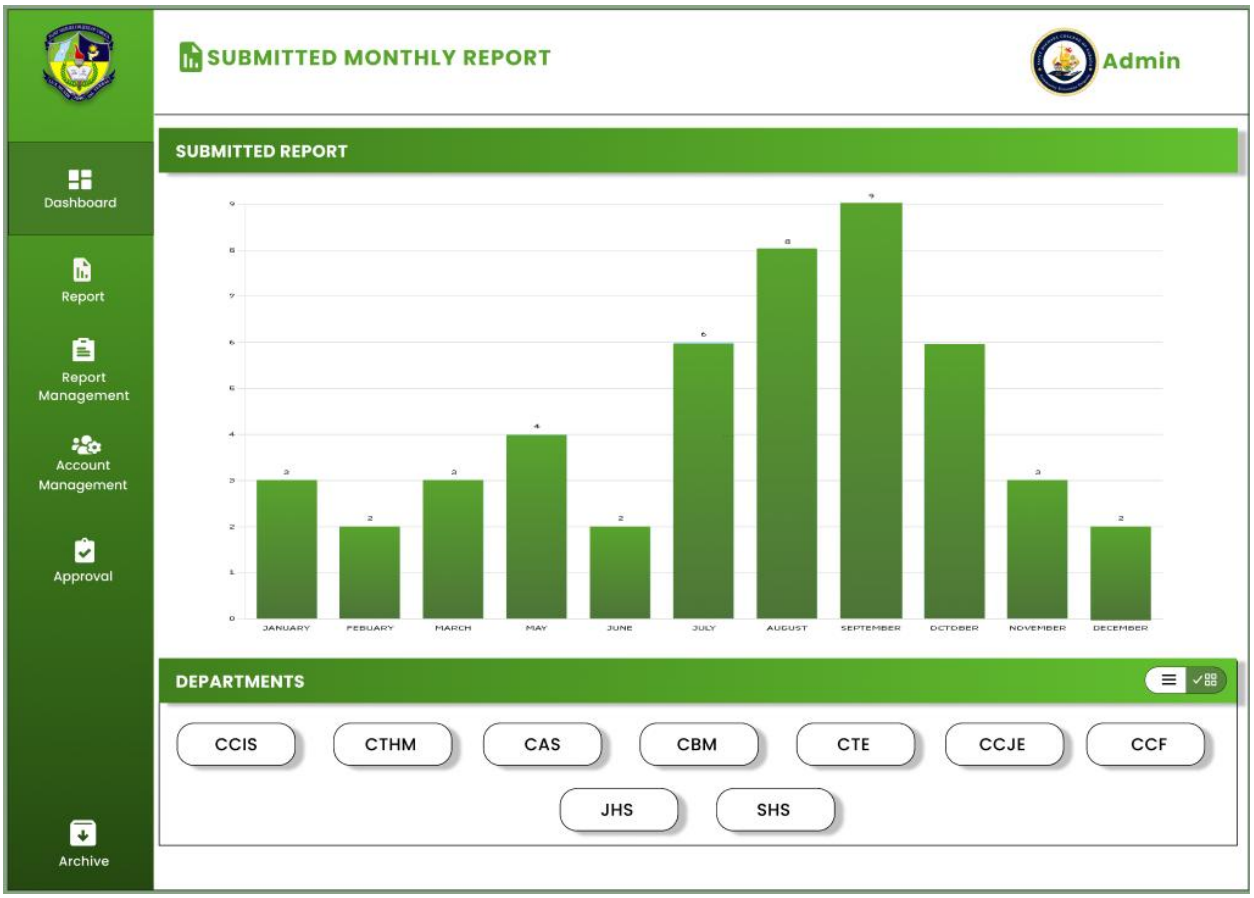


Figure 25 Submitted Monthly Report Dashboard

Figure 25 shows the admin's monthly report dashboard. Here, submission compliance can be easily tracked through a list of months and departments, so it can be quickly determined which units have or haven't submitted their reports at any given time.

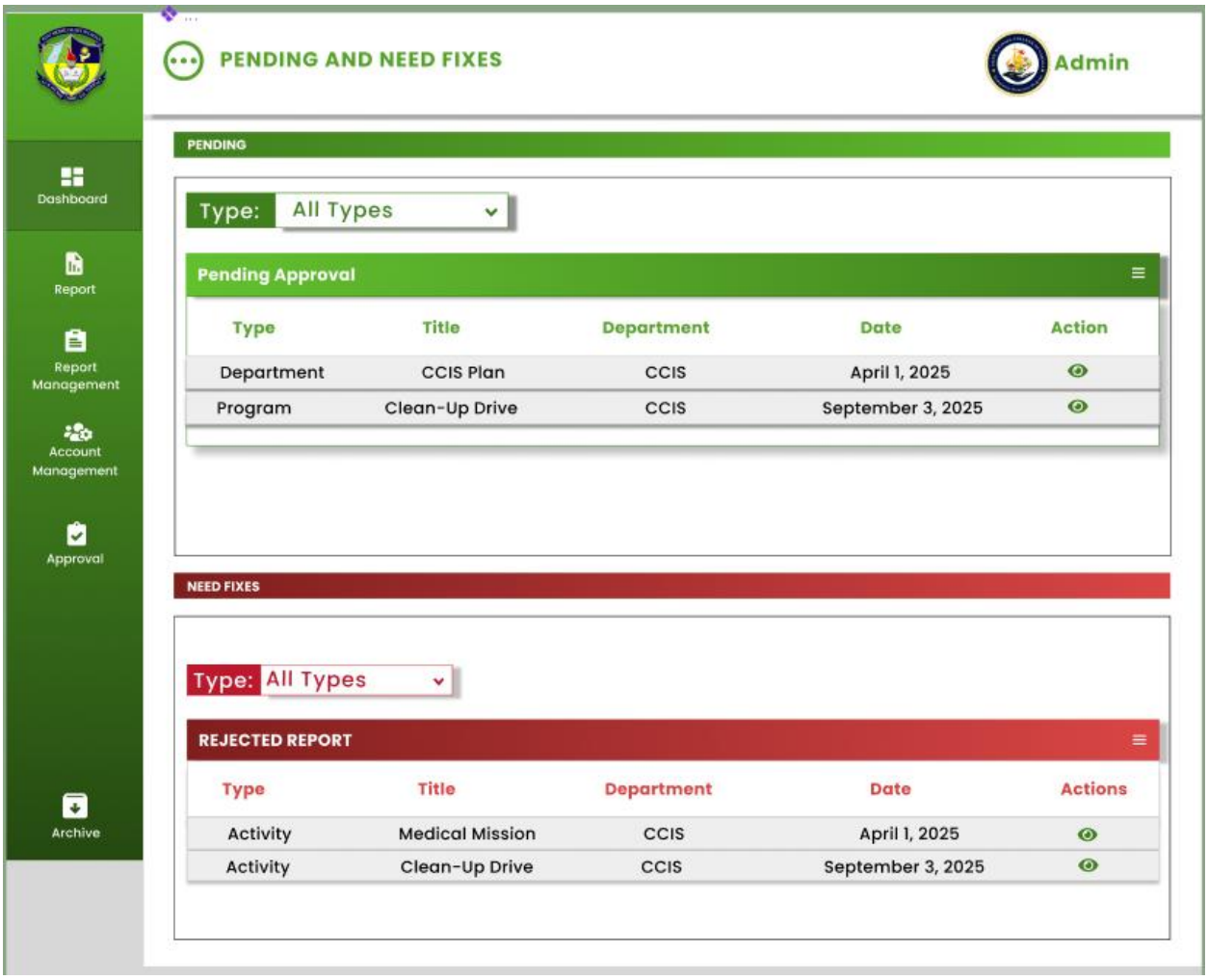


Figure 26 Admin Pending Reports Dashboard

Figure 26 shows the admin workflow dashboard divided into two parts. The “PENDING” tab contains reports awaiting initial review and approval, while the “NEED FIXES” tab is for rejected reports that need to be corrected and resubmitted by the coordinators. In this way, the admin can more easily manage the entire process of the review and correction cycle.

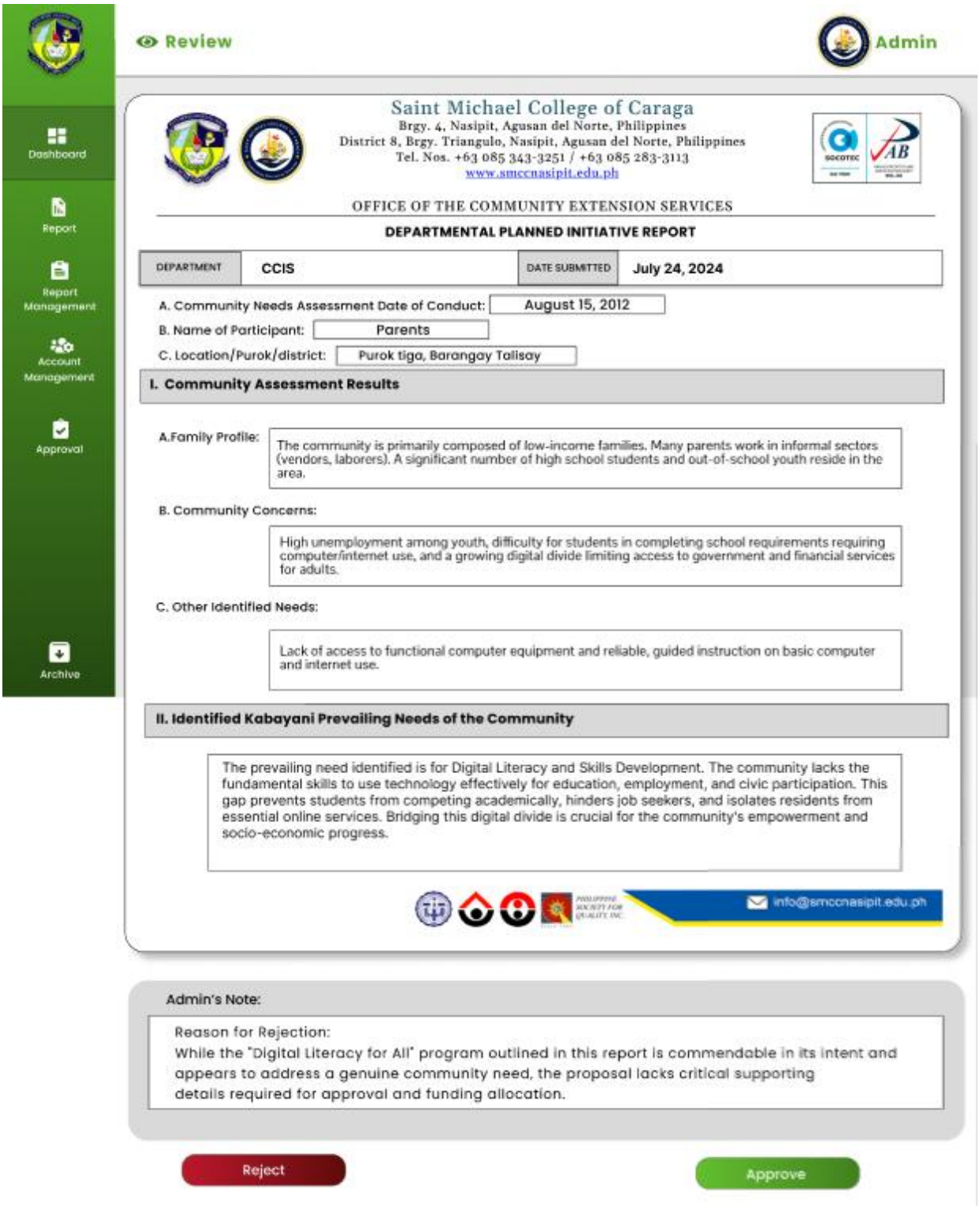


Figure 27 Admin Review

Figure 27 shows the admin review process for a submitted document. Once the department has submitted its Report, it will be posted on the dashboard page as pending for review by the administrator. Here, the administrator checks if the report details are complete, accurate, and correct. After the review, the Admin has two options: "Approve" if the Report passes, or "Reject" if something is missing or needs to be changed. When rejected, the Admin will enter the reason for rejection in the "Admin's Note" section, serving as a guide for the forwarder to correct the Report. Briefly, this page outlines the process for approving and providing administrative feedback on reports or projects submitted by departments.

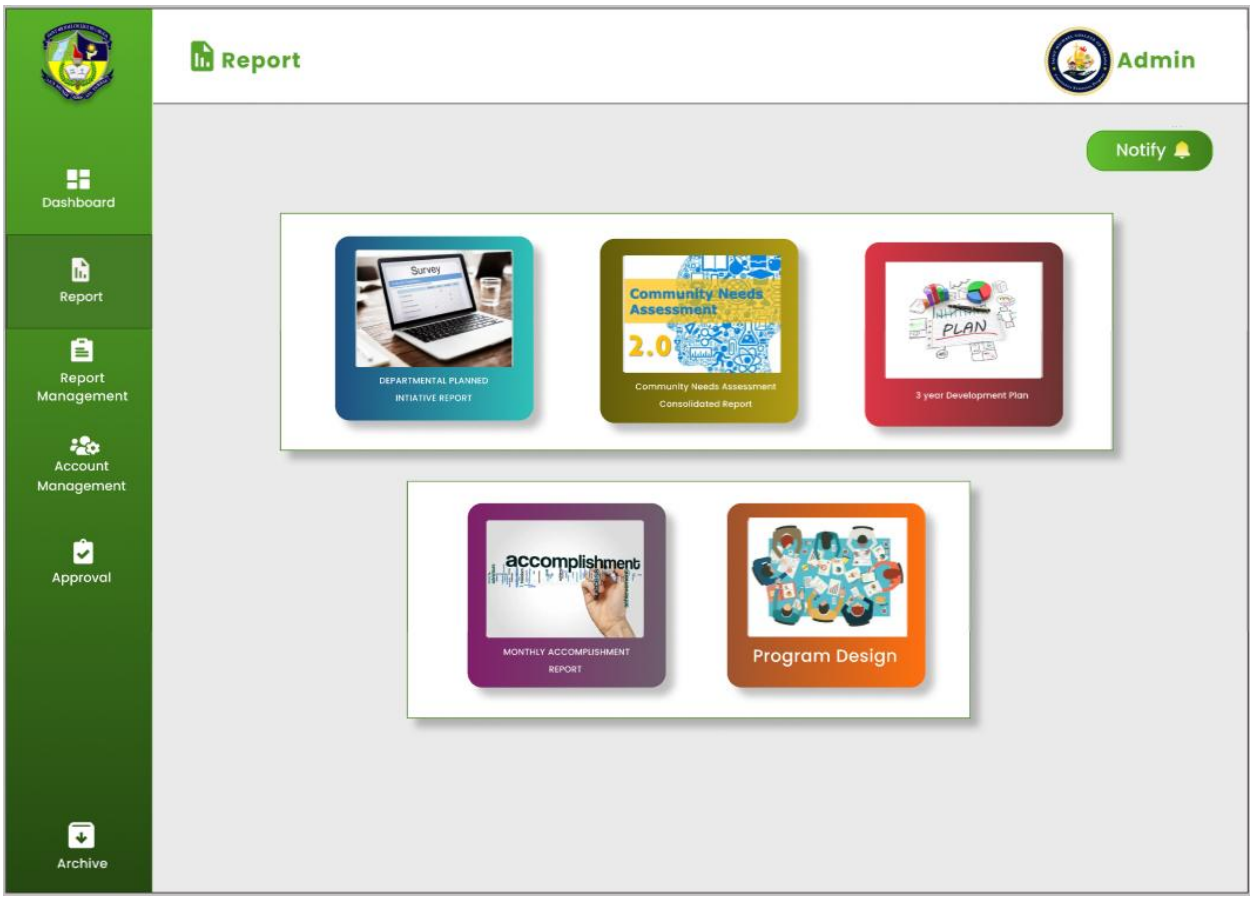


Figure 28 Report Navigation with Notification Button

Figure 28 shows another version of the Report that adds a notification button in the upper-right corner. This feature notifies the administrator whenever a new report or update is available from the

coordinators. The design of the reports remained the same colourful, neat, and clear to maintain an organized workflow. Each report type has a unique colour and picture that help identify the category easily, facilitating the management of each department's reports.

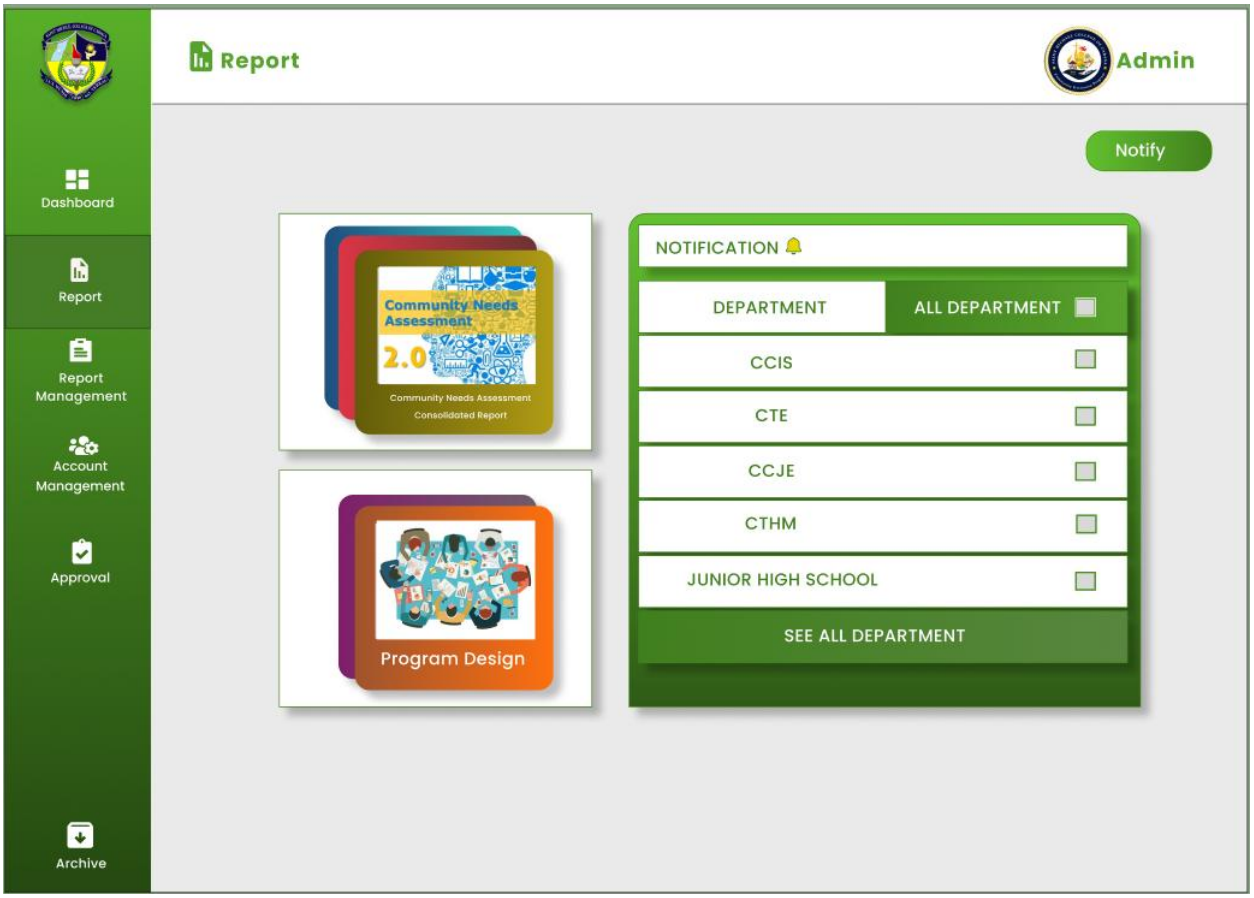


Figure 29 Notification for Coordinator Card in Report Navigation

Figure 29 shows a notification panel that opens when the Notify button is pressed. When pressed, a list of departments will appear. At the same time, five report cards will be compressed into two folders, one above and one below. The administrator may select one or more departments or select All Departments to send a reminder, a new report, or an important announcement. This feature helps speed up communication between the administrator and the coordinators. The design, featuring a combination of green and white, presents a professional and smooth appearance for managing notifications.

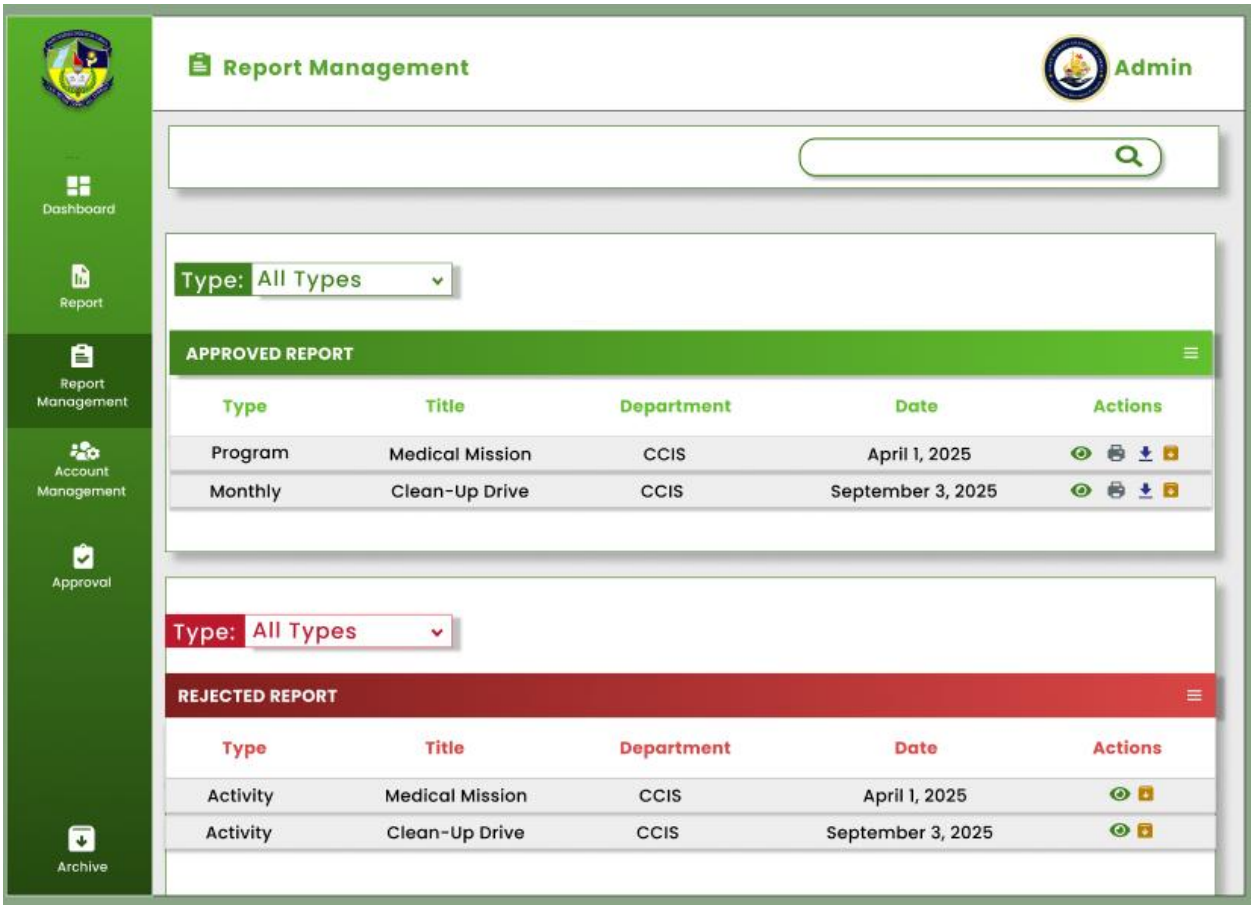


Figure 30 Report Management

Figure 30 shows the report management section of the administrator module. Here, all reports are organized into two categories: Approved Reports and Rejected Reports. Each table has columns for report type, title, department, submission date, and action buttons for viewing, printing, downloading, and archiving the report. There is also a dropdown filter for selecting the report type.

Green indicates approved reports and red indicates rejected reports, so the status of each report is easily visible. This helps administrators better manage and verify CES documents. Having an archive function allows systematic storage, to preserve data for future reference and auditing.

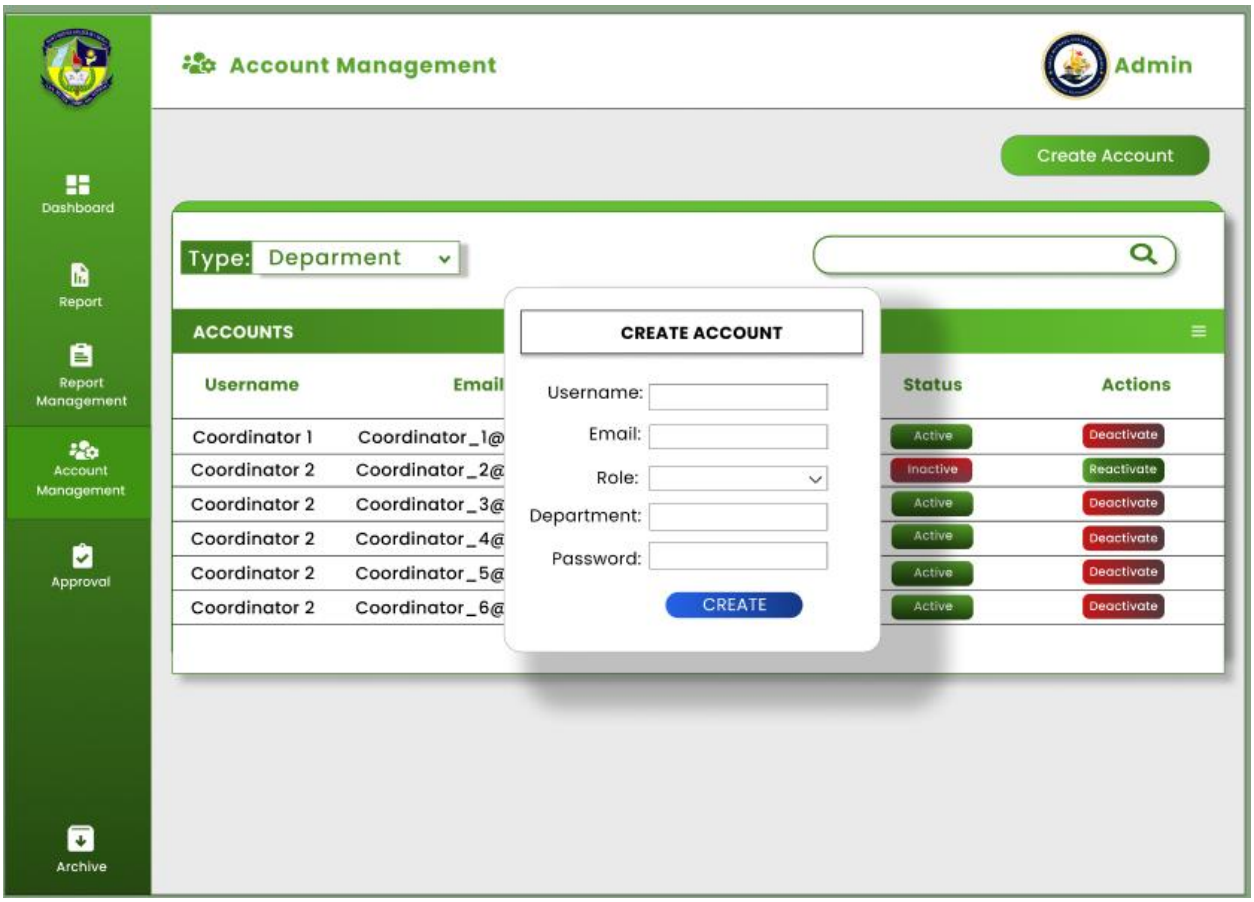


Figure 31 Account Management

Figure 31 illustrates the Account Management Interface, where administrators manage coordinators' accounts across various departments. Here is a list of Usernames, Email Addresses, Departments, and current account status. Active accounts are displayed in green, while inactive accounts are red. At the top right, there is a "Create Account" Button to create a new coordinator account, as well as a search bar to easily find specific users. Through this section, the security and organized management of users using the CES Workflow System is guaranteed.

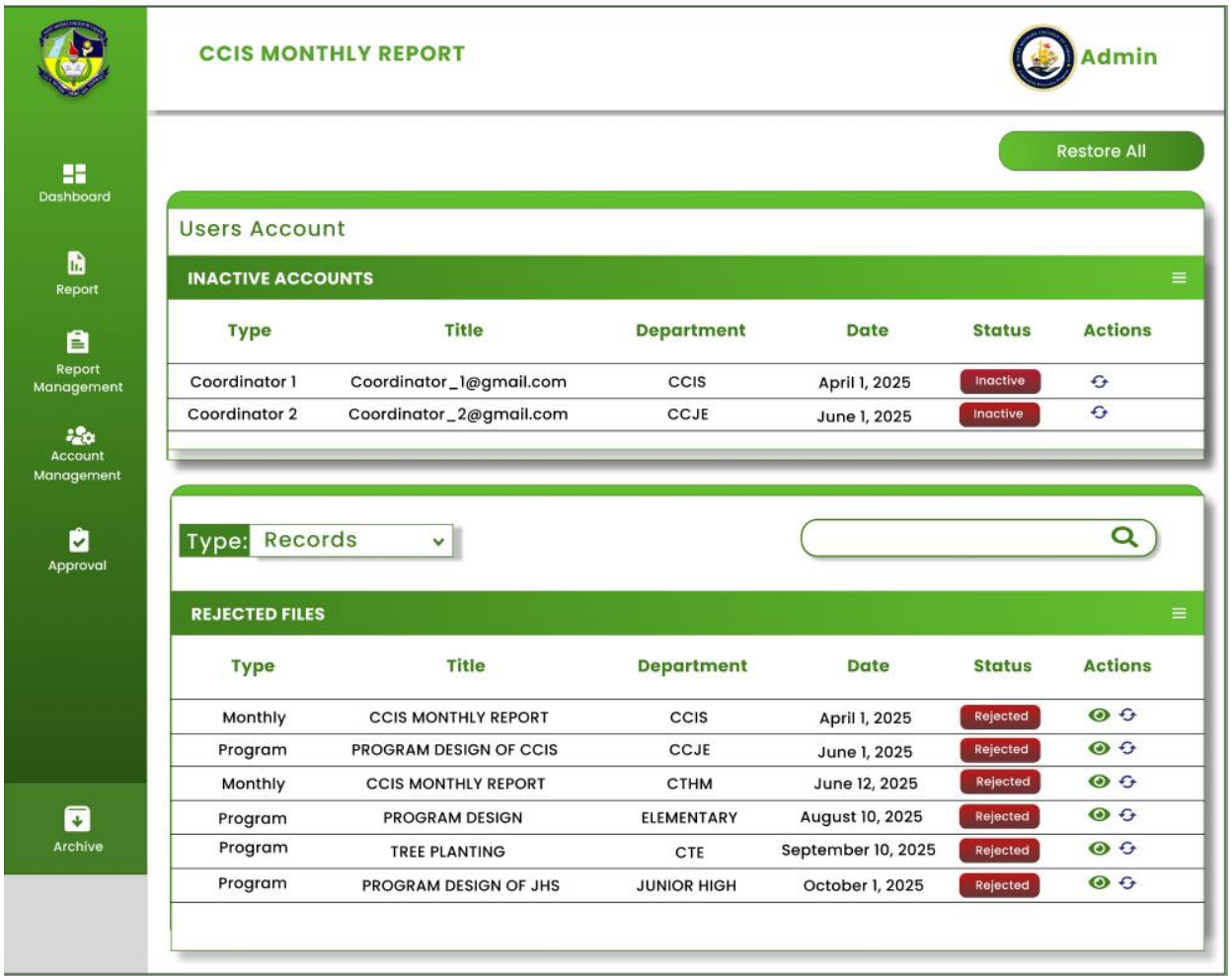
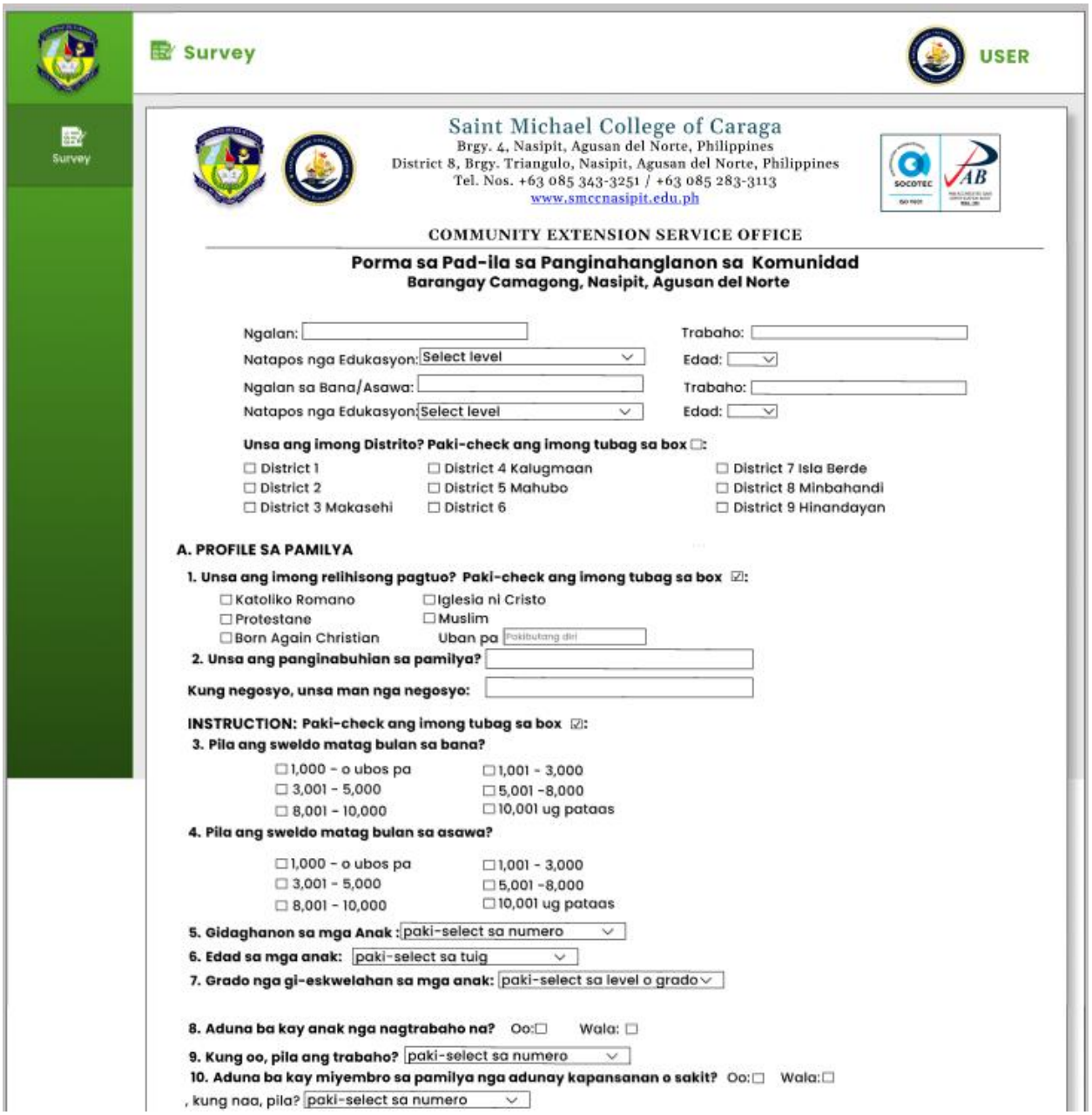


Figure 32 Admin Archive

Figure 32 shows the admin archive, where inactive accounts and rejected reports are located. For inactive accounts, admin can restore access. As for the rejected reports, the details can be viewed, and the reports returned. This ensures that all records are properly organized and easily accessible when needed.



Survey

USER

Survey

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COMMUNITY EXTENSION SERVICE OFFICE

Porma sa Pad-ila sa Panginahanglanon sa Komunidad
Barangay Camagong, Nasipit, Agusan del Norte

Ngalan: Trabaho:
Natapos nga Edukasyon: Select level Edad:
Ngalan sa Bana/Asawa: Trabaho:
Natapos nga Edukasyon: Select level Edad:

Unsa ang imong Distrito? Paki-check ang imong tubag sa box ☐:

☐ District 1 ☐ District 4 Kalugmaan ☐ District 7 Isla Berde
☐ District 2 ☐ District 5 Mahubo ☐ District 8 Minbahandi
☐ District 3 Makasehi ☐ District 6 ☐ District 9 Hinandayan

A. PROFILE SA PAMILYA

1. Unsa ang imong relihisong pagtuo? Paki-check ang imong tubag sa box ☑:

☐ Katoliko Romano ☐ Iglesia ni Cristo
☐ Protestane ☐ Muslim
☐ Born Again Christian Uban pa Pakibutang dri

2. Unsa ang panginabuhian sa pamilya?

Kung negosyo, unsa man nga negosyo:

INSTRUCTION: Paki-check ang imong tubag sa box ☑:

3. Pila ang sweldo matag bulan sa bana?

☐ 1,000 - o ubos pa ☐ 1,001 - 3,000
☐ 3,001 - 5,000 ☐ 5,001 -8,000
☐ 8,001 - 10,000 ☐ 10,001 ug pataas

4. Pila ang sweldo matag bulan sa asawa?

☐ 1,000 - o ubos pa ☐ 1,001 - 3,000
☐ 3,001 - 5,000 ☐ 5,001 -8,000
☐ 8,001 - 10,000 ☐ 10,001 ug pataas

5. Gidaghanon sa mga Anak : paki-select sa numero
6. Edad sa mga anak: paki-select sa tuig
7. Grado nga gi-eskwelahan sa mga anak: paki-select sa level o grado

8. Aduna ba kay anak nga nagtrabaho na? Oo:☐ Wala: ☐
9. Kung oo, pila ang trabaho? paki-select sa numero
10. Aduna ba kay miyembro sa pamilya nga adunay kapansanan o sakit? Oo:☐ Wala:☐
, kung naa, pila? paki-select sa numero

B. KABALAK-AN SA KOMUNIDAD

11. Unsa ang kasagaran nga problema sa panglawas sa inyong pamilya? Palihug hatagi ug numero 1-10. Ang numero 1 ang pinaka problema .

- ☐ Ubo ug Sip-on
- ☐ STD
- ☐ Diabetes
- ☐ Hilanat
- ☐ Migraine
- ☐ Dengue
- ☐ Diarrhea
- ☐ Tuberkulosis
- ☐ Typhoid Fever
- ☐ Rayuma
- ☐ High Blood Pressure

12. Uban pa, palihug tubagi: _____

13. Pagiaraw sa inyong kasilyas: Flush: ☐ Antipolo: ☐ Water-Sealed: ☐ Wala`y CR: ☐

14. Unsa ang inyong pamaagi sa paglabay sa basura: Compost Pit: ☐ Koleksyon sa Basura: ☐

15. Unsa ang pinaka-dali nga problema sa inyong distrito?

- ☐ Drug Trafficking
- ☐ Kalinaw ug Kahanas
- ☐ Malnutrition
- ☐ Kasilyas
- ☐ Early Pregnancy
- ☐ Panginabuhi
- ☐ Mga bata nga dili kamao mubasa
- ☐ Kakuwang sa trabaho
- ☐ Dugang kita para masuportahan ang pamilya

Uban pa, palihug tubagi:

16. Aduna ba mo`y problema sa kalinaw ug kahapsay sa inyong distrito? Oo: ☐ Wala: ☐

17. Kinahanglan ba ang training para sa mga barangay Tanod? Oo: ☐ Wala: ☐

18. Unsa man ang gikinahanglan nga mga training ang kinahanglan alang sa Barangay Tanod ug alang sa mga katawhan. Palihug hatagi og numero 1-10. Ang numero 1 ang pinaka problema.

- ☐ Basic Law Enforcement Training
- ☐ Basic Self-defense and Physical Fitness Training
- ☐ Community Policing and Conflict Resolution Training
- ☐ Crisis Intervention Training
- ☐ Disaster Preparedness and Response Training
- ☐ Weapons Handling and Security Procedures
- ☐ Human Rights and Ethical Conduct Training
- ☐ Environment Protection and Waste Mngt Training
- ☐ Fire Safety and Firefighting Training
- ☐ Cybersecurity and Online Safety Training

19. Unsa nga mga seminar ang pinakakinahanglan sa inyong distrito? Palihug hatagi ug numero 1-5. Ang numero 1 ang pinaka problema.

- ☐ Seminar bahin sa panglawas (HIV/STD, Drug Symposium, Early Pregnancy, mga sakit nga makatakod, sakit sa kasingkasing, rayuma, arthritis, diabetes, basic first aid)
- ☐ Seminar sa kalikupan (Pagreserba ug proteksyon sa kalipukan, hustong paglabay sa basura)
- ☐ Seminar sa relihiyon (Pagpalambo sa mga maayobatasan, responsable nga pagka-ginikanan)
- ☐ Edukasyon nga seminar (Crime Prevention, Basic Bookkeeping)
- ☐ Seminar sa panginabuhi (Organisasyon sa kooperatiba, pag-atiman og hayop)

Uban pa, palihug sa pagbutang: _____

20. Unsa nga mga training ang pinakakinahanglan sa inyong distrito? Palihug ranggohi;

- ☐ Baking ug cooking para sa mga inahan
- ☐ Computer Literacy training para sa out-of-school youth
- ☐ Crime Prevention ug disaster management para sa mga Barangay Tanod ug opisyales

Uban pa, palihug sa pagbutang: _____

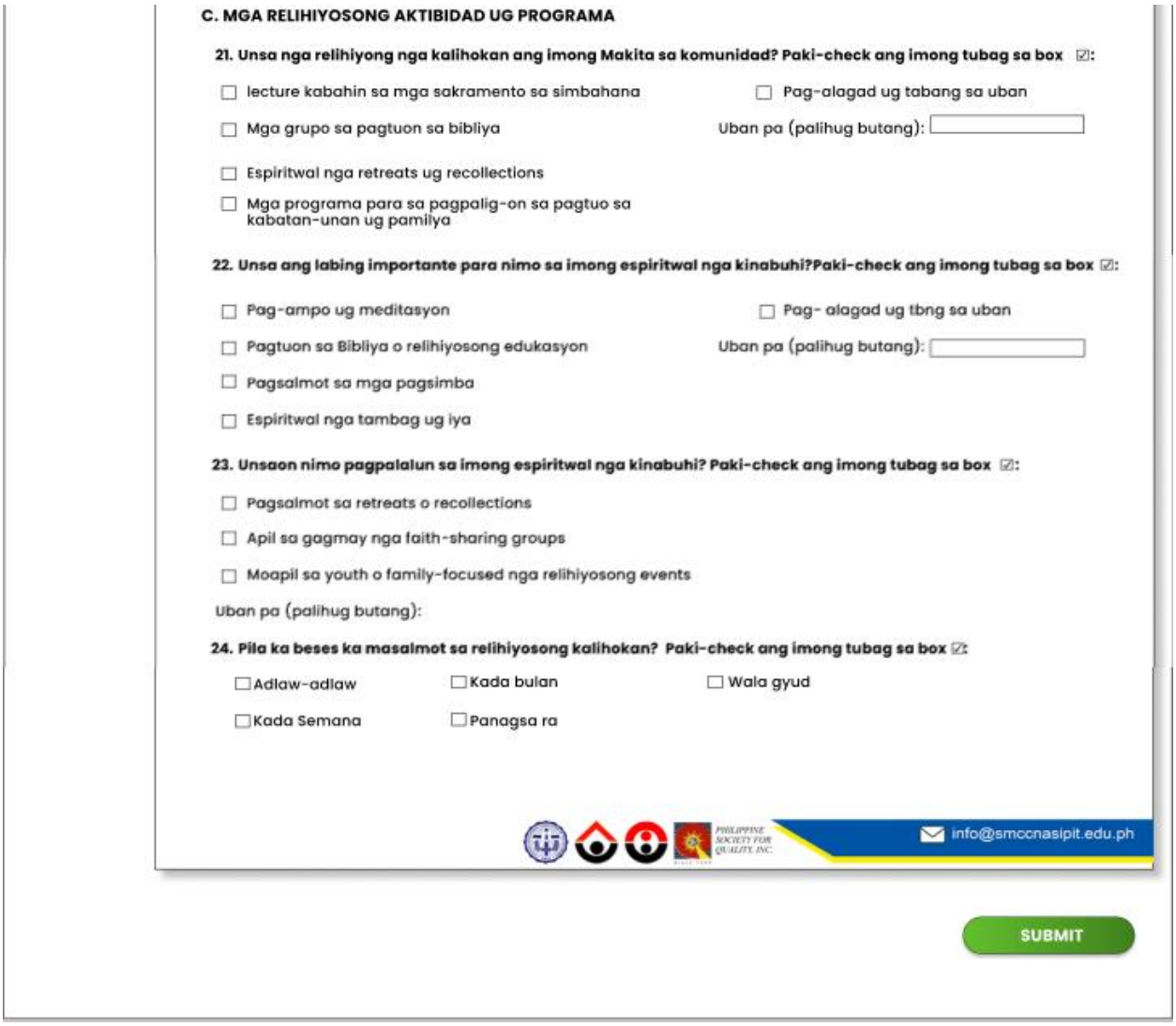


Figure 33 Field Survey Data Encoder

Figure 33 shows the Field Survey Data Encoder (FSDE) interface. After logging in, FSDE personnel use it to assist community members in completing demographic and needs assessment surveys. It collects important information of each household, such as family details, district, religion, livelihood, and monthly income, to ensure accurate and consistent data collection in the field.

Table 1

Software Platforms, Development Environments, and Tools

Components	Specification	Usage
Internet Browser	Any (Chrome, Edge, Firefox, etc.)	Users can access and view information from the internet with a web browser. After a successful login, the SMCC CES Workflow System interface loads, allowing users to access its functionalities.
Front-end	HTML	HTML specifies the structural elements of the SMCC CES Workplace System's interface, including forms, text, and other critical components.
	CSS	CSS is used to improve the aesthetics and design of HTML, creating a more engaging and responsive website.
	JavaScript	To make the website more interactive and provide users with more functionality, JavaScript handles all client-side functionality.
Backend	MySQL (Database)	Handles structured data storage and management. Through phpMyAdmin, developers can create tables, run queries, and maintain database integrity for the System.
	PHP	PHP is used on the server side to manage interactions between user input and the database, ensuring smooth data flow.
	Python	Supports AI-driven analysis and recommendation features by processing

		community data and generating insights that guide decision-making.
A.I Model & Tools	sentence-transformers (MiniLM) sklearn cosine similarity	Used to analyze the meanings of community needs and match similar projects or reports based on context. Used to retrieve past projects that resemble the current community needs for reference and inspiration.
Server	Apache	Using the XAMPP control panel, developers can manage the behaviour of its components, ensuring smooth, easy access to local host servers such as Apache and MySQL.

Table 2

Hardware Requirements

Components	Specification	Usage
Device	RAM (16 GB) Processor (Intel Core i5) SSD (128 GB) Flash Storage (8 GB)	This is like the computer's short-term memory. It helps the computer run many things at once and makes apps and files open faster. The main component of the computer—the processor, or CPU—is responsible for executing commands from the website. It also makes data processing easier. This is where the computer saves programs, reports, and files. It works faster than old hard drives, so you can open and save things quickly. These storage solutions ensure quick retrieval and secure long-term data handling.
Printer	Any Ink Jet Printer Units	Inkjet printers produce clear, high-quality output used to print compliance documentation, data summaries, and audit reports. Since these printers are reasonably priced and multipurpose, they can be used for routine audit management printing, including creating formal documents for internal use or sending them to stakeholders.

System Evaluation

The evaluation of the SMCC CES Workflow System with AI Analysis and Recommendation was conducted using two approaches: the ISO 25010 quality model and AI/ML evaluation metrics. This dual framework ensures that the System is not only functional, efficient, and usable but also that its AI component delivers accurate and reliable recommendations. *Data was analyzed and interpreted based on the following parameters:*

Table 3

Adjectival Rating	Scale Range (Mean)	Verbal Interpretation
4	3.50 – 4.0	Very Functional
3	2.50 – 3.49	Functional
2	1.50 – 2.49	Moderately Functional
1	1.0 – 1.49	Poor Functional

Functional Suitability

Table 3 presents the criteria used to evaluate a product or System's functional suitability according to the ISO survey.

This measurement highlights the degree to which the System's modules—such as reports and AI analysis—align with user expectations and deliver the intended outcomes.

Table 4

Performance Efficiency

Adjectival Rating	Scale Range (Mean)	Verbal Interpretation
4	3.50 – 4.0	Very Efficient
3	2.50 – 3.49	Efficient
2	1.50 – 2.49	Moderately Efficient
1	1.0 – 1.49	Poor Efficient

Table 4 outlines the scoring parameters used to assess the System's overall performance efficiency in executing its functions and tasks.

Table 5

Usability

Adjectival Rating	Scale Range (Mean)	Verbal Interpretation
4	3.50 – 4.0	Very Usable
3	2.50 – 3.49	Usable
2	1.50 – 2.49	Moderately Usable

1	1.0 – 1.49	Poor Usable

Table 5 presents the scoring parameters used to assess the System's usability

Table 6

AI/ML Evaluation Metrics

Metrics	Description	Purpose in System Evaluation
Accuracy	Proportion of correct predictions made by the AI model.	Ensures AI recommendations match real community needs.
Precision	Proportion of true positives among all predicted positives.	Confirm that AI suggestions are relevant and not misleading.
Recall (Sensitivity)	Proportion of true positives among all actual positives.	Ensure important data patterns or needs are not overlooked.
F1-Score	The harmonic means of precision and recall provides a balanced measure, especially useful for imbalanced classes.	Useful when data classes are imbalanced to show overall effectiveness
Confusion Matrix	A table shows true positives, true negatives, false positives, and false negatives.	Provides a detailed view of AI's performance and classification accuracy.

AUC-ROC	Area under the Receiver Operating Characteristic curve.	Show how well the AI separates different classes (good vs bad cases).
Mean Absolute Error (MAE)	Average of the absolute differences between predicted and actual values.	Measures how close AI predictions are to fundamental values, in simple units.
Root Mean Square Error (RMSE)	Square root of the average squared differences between predicted and actual values.	Highlight bigger errors more strongly, useful for continuous predictions.
User Satisfaction	Feedback from system users on the usefulness and clarity of AI outputs.	Validates whether AI recommendations are understandable and beneficial in real decision-making.

Table 6 presents the AI/ML evaluation metrics used to assess the accuracy, reliability, and usefulness of the System's recommendation engine.

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Weight: 42kg
Civil Status: Single

Educational Attainment

Elementary: Manapa 2 Central Elementary School
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Tertiary: Saint Michael College of Caraga

CURRICULUM VITAE



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Sex: Female
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Weight: 52kg
Civil Status: Single

Educational Attainment

Elementary: Camagong Elementary School
Secondary: Saint Michael College of Caraga
Tertiary: Saint Michael College of Caraga

RESEARCH ETHICAL STANDARD

A. Protection of Intellectual Property Rights (IPR)

The capstone project ensures that all intellectual rights related to the CES Workflow System with AI analysis and recommendations for Saint Michael College of Caraga are protected. The study expressly acknowledges that the system code, database design, UI layout, AI algorithm, and entire documentation are original creations of the researchers, so it may be subject to the protection of Republic Act No. 8293 or Intellectual Property Code of the Philippines. Since this project contains original software and an AI-based recommendation model, the study also recognizes the possibility of it being patentable depending on the evaluation of the Intellectual Property Office of the Philippines (IPOPHIL).

In addition, all parts of the system such as source code, UI design, database schema, and documentation are covered by copyright as original literary and artistic works. The study also ensures that all third-party tools, frameworks, and libraries used, such as Bootstrap, SentenceTransformer, and Cosine Similarity, have proper compliance with their respective licenses and have clear acknowledgment or citation. Researchers used only open-source or legal methods permitted by licenses, and all sources were properly cited in the documentation. Since no unique brand name or logo has been created for this system, no trademark or application has yet been claimed for it.

B. Informed Consent

A clear and comprehensive explanation was given to all study participants including CES staff, coordinators, and other involvements about the purpose, scope, and overall process of developing and evaluating the CES Workflow System with AI analysis and recommendation. They are explained what data will be collected, such as login credentials for testing, reports or observations while using the

System, and any feedback related to its performance and functionality. The researchers also highlighted possible risks, such as exposure or misuse of data, and benefits, such as a more efficient reporting system and easier workflow at CES.

Before conducting any data collection, the researchers asked for a written or digital consent form to ensure that each participant's participation was voluntary. They were also reminded that they have the right to withdraw from the study at any time without any negative impact on their work or relationship with the institution. All consent forms were properly recorded and kept as part of the documentation in compliance with ethical research standards.

C. Data Privacy and Confidentiality

Participation in the development and evaluation of the SMCC Community Extension Service (CES) Workflow System with AI Analysis and Recommendation is completely voluntary, but because the system handles personal data such as name, email address, login credentials, reports, and community-related information, it is covered by the Data Privacy Act of 2012 (RA 10173). Participants were told that sensitive data is encrypted and stored securely in the database. Only authorized personnel, specifically the system administrator, have access to the encrypted data and control panel, while regular users have limited access based on the Role-Based Access Control (RBAC) implemented in the system.

In order to protect the identity of participants, any data used for analysis was subjected to an anonymization process, where personal identifiers were removed, edited, or replaced with coded references. The researchers also ensured that no data would be shared, displayed, or released without the consent of the participants and strictly followed the confidentiality protocols of the institution. There remained no force, influence, or any compensation used to obtain their participation, and the researchers kept the entire process fair, just, and voluntary.

D. Voluntary Participation and Freedom to Withdraw

Participation in the development and testing of the SMCC Community Extension Service (CES) Workflow System with AI Analysis and Recommendation is strictly voluntary. Although CES coordinators are institutional members, they are under no obligation to take part in the system testing or provide feedback. They are clearly informed that choosing not to participate or deciding to withdraw at any stage will not result in any academic, professional, or institutional consequences.

There is no use of coercion, influence, or any form of financial rewards to obtain their participation. In this way, researchers have made sure that all participation is fair, impartial, and by genuine volunteers.

E. Minimization of Harm and Risk Management

A risk assessment was conducted to ensure that no physical, psychological, or professional harm would affect the study participants. While the project poses only minimal risk, potential concerns such as data security issues and workload-related stress may still arise. To mitigate these risks, the system employs secure password protection and controlled access to safeguard participant data. Debriefing sessions are also provided to address concerns, clarify procedures, and reduce stress during system use. In addition, backup protocols are in place to prevent data loss and ensure smooth, uninterrupted participation throughout the evaluation process.

Participants' comments and responses are subject to the anonymization process to prevent any negative impact on their professional status. Researchers also provided assistance and support in the event of any problems during the project.

Since no animals were used or included in the study, the 3Rs principle (replacement, reduction, refinement) does not apply to this project.

F. Beneficence and Contribution to Knowledge

This his Capstone Project aims to provide an effective and technology-supported system for the Community Extension Service (CES) Department of Saint Michael College of Caraga to enhance their workflow, documentation, and management of community service projects. Through the SMCC Community Extension Service (CES) Workflow System with AI Analysis and Recommendation, the department can streamline processes such as project proposal submission, evaluation, approval, monitoring, and report generation. These improvements are expected to make the handling of reports, project approvals, and documentation faster, more organized, more transparent, and more sustainable.

The project directly benefits the CES Department by improving the efficiency and reliability of report handling and strengthening the overall transparency of extension-related processes. It also contributes to broader knowledge in the fields of technology, information systems, and community engagement by demonstrating how AI-assisted workflow systems can support, optimize, and enhance institutional operations. This project serves as a model of how academic institutions can integrate AI into community extension workflows to improve decision-making and project alignment with community needs.

The findings of the study including system performance, user feedback, AI-generated insights, and evaluation results will be formally shared with the SMCC CES Department to inform them of how the system supports their operational needs. Additionally, any system improvements, enhancements, or refinements developed during or after the study will also be provided to the CES Department. These may include optimized features, interface enhancements, improved recommendation accuracy, or additional modules added as part of the system's continuous development. A summary of the results may also be shared with participants upon request to maintain transparency and keep stakeholders informed of the project's outcomes.

G. Justice and Fair Participant Selection

The selection of participants for this Capstone Project was performed fairly and impartially. Participants were identified as Department Coordinators and CES staff at Saint Michael College of Caraga. They were chosen because their job duties are directly relevant to the system being reviewed and they actively use the institutional activities, making them the most qualified and knowledgeable sources of data. The selection of participants was based on this job relevance and their willingness to participate in system reviews and utilization, and eliminating any potential for personal preference or bias in the recruitment process. This targeted approach ensured that inclusion was equitable, meaning all staff whose professional roles required interaction with the system were given the opportunity to participate, and the process was strictly non-discriminatory, avoiding the exclusion of any group based on factors other than professional involvement. Researchers confirm that the inclusion was equitable and non-discriminatory, ensuring no group was unfairly burdened, depressed, or overly represented to maintain a fair and clear participant selection process. In this way, the principle of justice and equality has been protected throughout the project's entire work.

H. Data Integrity and Accuracy

All data for developing and reviewing the SMCC CES Workflow System with AI Analysis and Recommendation are compiled using standardized methods to ensure accuracy and reliability. The researchers conducted a validation process that includes manual verification, where AI-generated outputs are carefully compared with the original inputs and project requirements. This involves checking whether the AI correctly interprets the community need, whether the recommended projects remain relevant to the context, and whether the similarity results reflect appropriate levels of relevance. Any inconsistencies or irrelevant suggestions identified through this review are noted and corrected.

To maintain the quality and accuracy of the system, the researchers also performed general debugging actions. These include examining system behavior when unexpected results appear, reviewing system messages or logs to identify potential sources of error, testing multiple inputs to check for inconsistencies, and adjusting system settings or processes when outputs do not align with expected results. These actions ensure that errors are detected, traced, and resolved throughout system development.

AI tools such as Sentence Transformers (MiniLM) and Cosine Similarity were used to analyze the semantic meaning of community needs and match them with similar previous projects. The AI system generates recommendations that align with the nature and context of each unit or area, ensuring relevance and appropriateness of the suggested outreach activities.

Possible biases, errors, and limitations—such as sampling method, variations in input quality, or model interpretation—are recorded and considered. Bias mitigation strategies were applied to minimize inaccurate outputs. All AI-generated data undergo thorough human verification to maintain the integrity and reliability of the study.

I. Transparency and Honesty in Reporting

The findings and result of the study and evaluation of the SMCC Community Extension Service (CES) Workflow System with AI Analysis and Recommendation will be reported fully, honesty , completely, impartially. The researchers guarantee that no data will be manipulated, altered, or selectively presented, and that all significant outcomes whether favourable, unfavourable, or unexpected will be clearly reflected in the final report. The researchers alco commit to the truthful reporting of all system test results and identified limitations, as well as the proper acknowledgment of all third-party resources used in the conduct of this study.

J. Use of Patented or Copyrighted Materials

All data related to the design, review , and development of the SMCC Community Extension Service (CES) Workflow System with AI Analysis and Recommendation are carefully gathered to ensure accuracy and integrity. Researchers adhered the standard data collection and documentation procedures to ensure that comments, observations, and evaluation results were accurately recorded and interpreted. During the system development process, API service were used to obtain and process data from third-party platforms and all API calls were preformed in accordance with the license and terms of use provided by the service providers, ensuring that no sensitive information was accessed or released without permission. The system also uses several third-party components (e.g., Bootstrap, PHP frameworks, Sentence Transformer, and MySQL), all which are open-source tools that are properly acknowledged and attributed in accordance with their respective licenses. The researchers further affirm that no proprietary or patented materials were used without proper authorization.

K. Ethical Consideration for Animals and Human Trials

The project includes participants from the Primary, Secondary, and Higher Education departments, as well as the CES Department. All participants are fully informed of their rights, the low levels of risk, and the benefits of their project participation through a formal Informed Consent process. We offer explicit assurance that their rights and well-being will be respected, and that participation will cause no harm, as the study activities involve only minimal risk associated with providing professional feedback. Participation is entirely voluntary, and staff are informed of their right to withdraw at any time without penalty or prejudice to their professional status.

Ethical clearance for animal use is inappropriate because there is no animal involved in the study. All professional feedback or feedback provided by the participants is subject to the anonymization process to ensure that there is no negative impact on their professional status or reputation.

M. Ethical Clearance and Institutional Approval

This capstone project , which was carefully designed and developed to adhere to the highest standards of academic integrity and research excellence, successfully received ethical approval from Institutional Review Board of Saint Michael College of Caraga, with the ethical clearance being formally granted after a thorough review process that ensured full compliance with the institution's established guidelines and protocols concerning data privacy and confidentiality, the protection of intellectual property rights, and the responsible and ethical treatment of human participant who were involved in the study.

L. Responsible Use of AI and Other Related Technologies

The researchers utilized AI tools to expedite the search for relevant literature (RRL) in Google Scholar and Mendeley. Grammarly was also used to ensure proper grammar, a clear flow of ideas, and proper expression of the paper. The researchers ensured that the use of AI was responsible and with integrity, by analyzing and validating all the information obtained. AI was used only as an aid, and the final decisions were made by the researchers to maintain the ethical standards of the study. The project has been approved by the Saint Michael College of Caraga (SMCC) Research Ethics Committee (REC) before the development of the system.